COMSATS University Islamabad Wah Campus



Department of Computer Science

Internship Report (Week No. 2)

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Chapter 1: Introduction

1.1 Introduction

During the second week of my internship at **Pakistan Ordnance Factories (POF)**, I gained practical knowledge of Oracle Database Administration by performing advanced database installation and configuration tasks. The primary focus of this week's training was to manually install Oracle Database 12c on Oracle Linux 7.9 running in VMware Workstation. Unlike automatic installation methods, the manual installation process required proper Linux configuration, execution of terminal commands, and Oracle software setup using Oracle Universal Installer (OUI).

After completing the database installation, I learned how to configure the Oracle environment after every system reboot by setting the required environment variables, starting the Oracle Net Listener service, and opening the Oracle Database using SQL*Plus. These tasks are essential for a Database Administrator (DBA) because the Oracle services must be started correctly before users or applications can access the database.

In the final practical of this week, Oracle SQL Developer was installed on Microsoft Windows, and a successful connection was established with the Oracle Database running on Oracle Linux. SQL Developer provides a graphical interface for executing SQL queries, creating database objects, and performing database management tasks. This practical prepared the environment for future database development and the implementation of the internship project.

Overall, the second week provided practical experience in Oracle Database installation, Linux-based database administration, and client-server connectivity, which are fundamental skills required for an Oracle Database Administrator.

1.2 Objectives

The objectives of this week's practical training were:

- To understand the manual installation process of Oracle Database 12c on Oracle Linux 7.9.
- To configure Oracle Linux for Oracle Database installation.
- To learn the Oracle Universal Installer (OUI) installation procedure.
- To execute the required root scripts after installation.
- To create and configure an Oracle database using the Database Configuration Assistant (DBCA).
- To verify the successful installation of Oracle Database using SQL*Plus.
- To learn how to configure Oracle environment variables after restarting the Linux system.
- To start and verify the Oracle Net Listener service.
- To start and open the Oracle Database after system reboot.
- To install Oracle SQL Developer on Windows.
- To establish a successful connection between Oracle SQL Developer and Oracle Database 12c.
- To prepare the database environment for future SQL development and database administration tasks.

1.3 Software and Hardware Requirements

The following software and hardware were used during this week's practical training.

Component	Description
Virtualization Software	VMware Workstation
Operating System	Oracle Linux 7.9
Database Software	Oracle Database 12c Release 1 (12.1.0.2)
Database Client	Oracle SQL Developer
Database Installation Files	database1.zip and database2.zip
Terminal	Oracle Linux Terminal
RAM	Minimum 8 GB
Processor	Intel Core i5 or Higher
Hard Disk	Minimum 100 GB Free Space
Network	Local Network for Database Connectivity

Chapter No 2 Oracle Database 12c Installation Using Local YUM Repository on Oracle Linux 7.9 in VMware

2.1 Objective

The objective of this practical is to install Oracle Database 12c on Oracle Linux 7.9 using the **Local YUM Repository Method**. This method prepares the operating system by configuring a local software repository, installing Oracle prerequisite packages, creating the required Oracle environment, and launching the Oracle Universal Installer. This approach simplifies the installation process by automatically installing the necessary dependencies and creating the required Oracle user and groups.

2.2 Requirements

Before starting the installation, ensure that the following software, files, and resources are available.

Software Requirements

- VMware Workstation
- Oracle Linux 7.9 Virtual Machine
- Oracle Linux 7.9 Installation DVD/ISO
- Oracle Database 12c Installation Files (Part 1 & Part 2)
- Local YUM Repository
- Shared Folder (DB for Linux)

User Requirements

- Root User Account
- Oracle User (Created Automatically by Oracle Pre-installation Package)

System Requirements

- Minimum 4 GB RAM
- At least 40 GB Free Disk Space
- Internet Connection (Optional)

2.3 Configuring Local YUM Repository

Introduction

A **YUM (Yellowdog Updater Modified) Repository** is used to install software packages and their dependencies in Oracle Linux. Instead of downloading packages from the internet, this practical uses the **Oracle Linux Installation DVD** as a **Local Repository**. This method is faster, does not require an internet connection, and is commonly used in enterprise environments.
The following steps explain how to configure the Local YUM Repository.

Step 1: Login as Root User

Explanation

Power on the Oracle Linux virtual machine and log in using the **root** account. Root privileges are required because repository configuration and software package installation are administrative tasks.

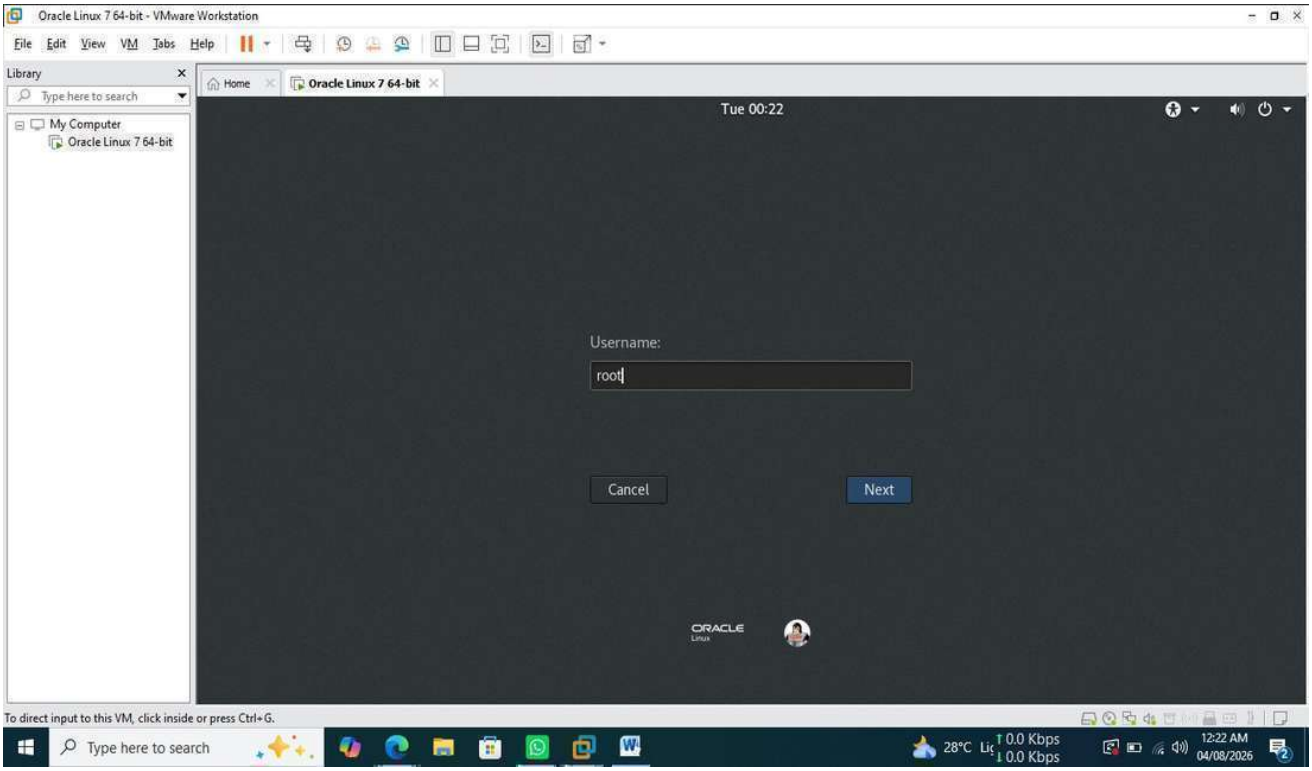
Verification Command

whoami

Expected Output

Root

Figure 2.1: Login as Root User



Step 2: Verify Oracle Linux Installation Media

Explanation

After logging in, verify that the Oracle Linux Installation DVD/ISO is successfully mounted. The mounted DVD contains the **Packages** directory, which will be used to create the Local YUM Repository.

Command

df -h

or

mount | grep sr0

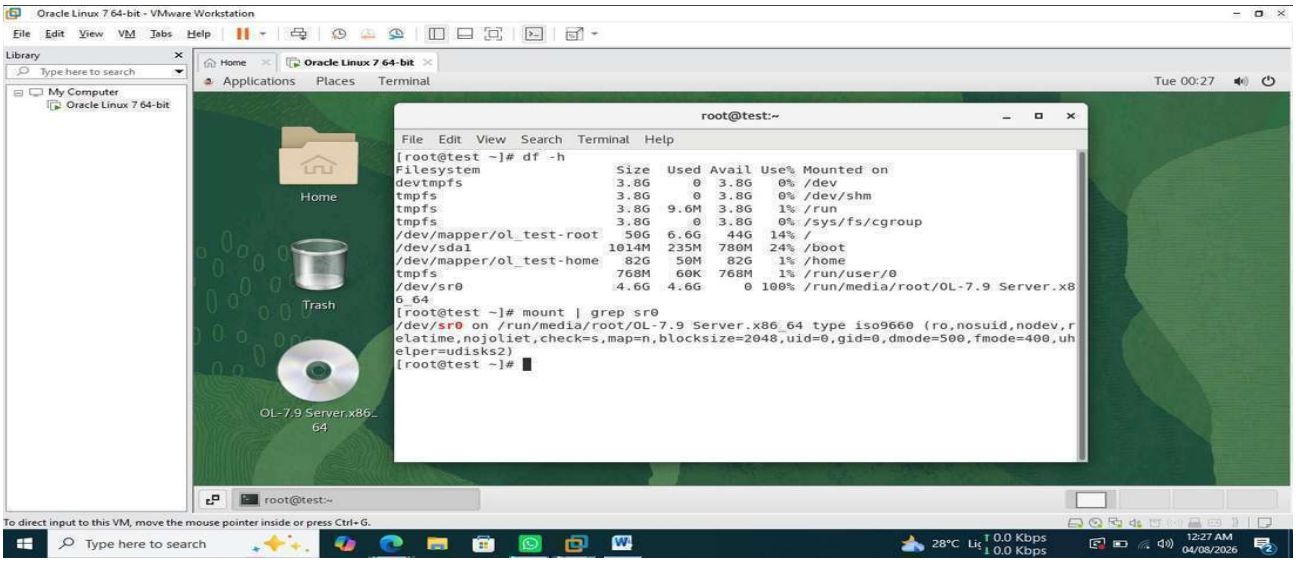
Command Explanation

- **df -h** → Displays mounted file systems.
- **mount | grep sr0** → Displays the mounted DVD device.

Expected Result

The mounted Oracle Linux DVD should appear in the output.

Figure 2.2: Oracle Linux Installation Media



Step 3: Create Local Repository Directory

Explanation

Create a directory named **repo** in the root (/) directory. This directory will store the package files copied from the Oracle Linux installation media.

Command

mkdir /repo

Command Explanation

- **mkdir** → Creates a new directory.
- **/repo** → Local repository directory.

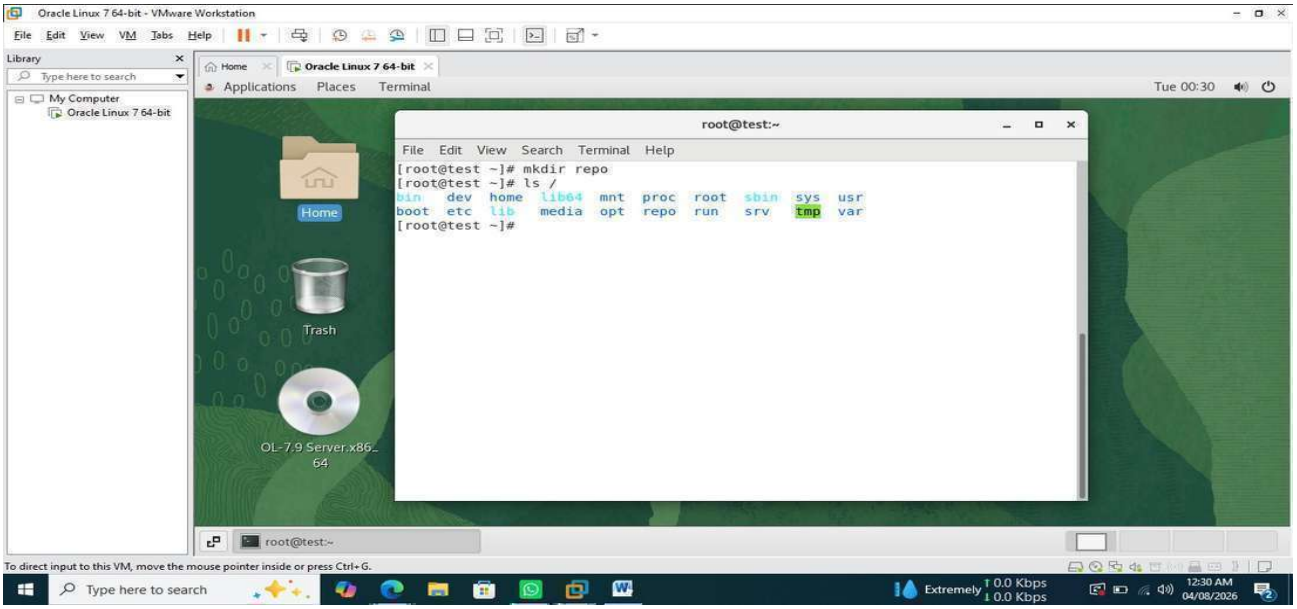
Verification Command

ls /

Expected Output

The **repo** directory should be visible in the root directory listing.

Figure 2.3: Creating Local Repository Directory



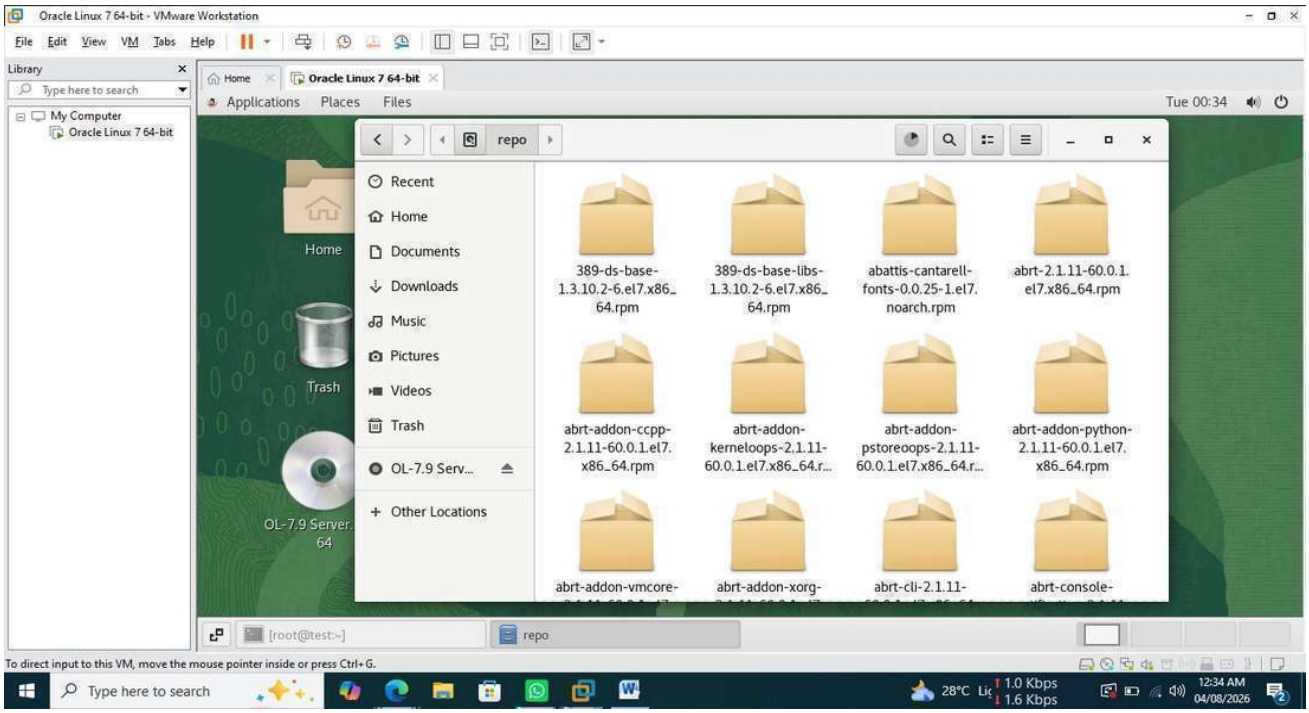
Step 4: Copy the Packages Folder

Explanation

Open the mounted Oracle Linux DVD from the desktop. Locate the **Packages** folder and copy all package files into the **/repo** directory. These packages will be used to build the Local YUM Repository.

Note: In our lab, the packages were copied using the graphical file manager.

Figure 2.4: Copying Oracle Linux Packages to /repo



Step 5: Configure the Local YUM Repository

Explanation

After copying all Oracle Linux packages into the **/repo** directory, the next step is to configure a **Local YUM Repository**. Oracle Linux stores repository configurations inside the /etc/yum.repos.d/ directory. In this practical, the existing repository configuration is modified so that YUM installs packages from the local repository instead of the internet.

Command

cd /etc/yum.repos.d

ls

Open the repository configuration file:

vi public-yum-ol7.repo

Repository Configuration

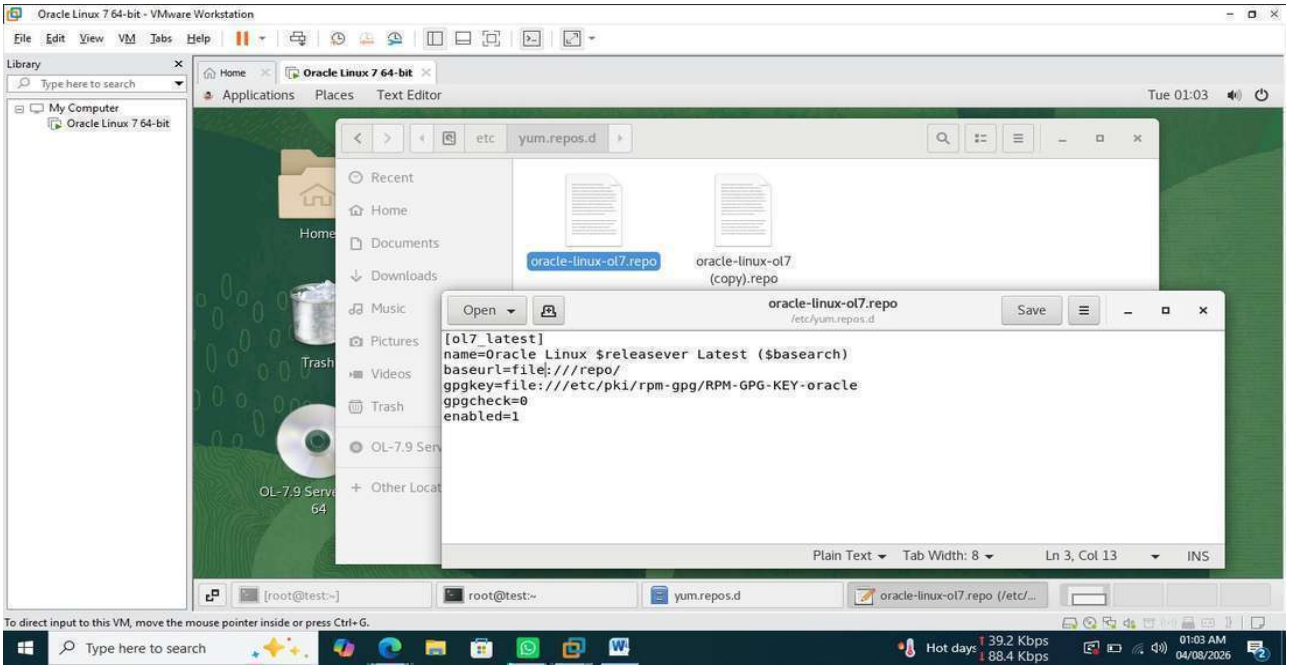
Replace or modify the repository configuration according to your local repository.

Note: Use the same repository configuration that your instructor demonstrated in the lab.

Command Explanation

- cd /etc/yum.repos.d → Opens the YUM repository configuration directory.
- ls → Lists all available repository files.
- vi public-yum-ol7.repo → Opens the repository configuration file for editing.

Figure 2.5: Configuring the Local YUM Repository



Step 6: Clean Existing YUM Cache

Explanation

Before using the newly configured repository, remove the old cached repository information. This ensures that Oracle Linux reads the updated repository configuration.

Command

yum clean all

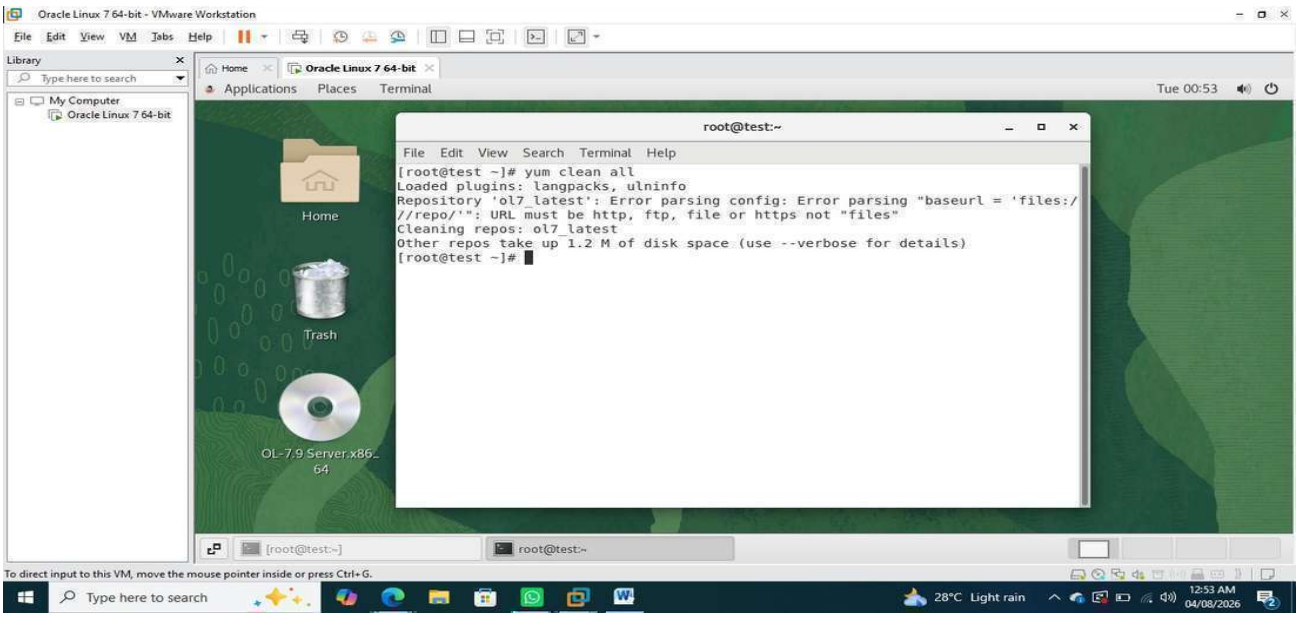
Command Explanation

- yum clean all → Removes all cached repository data.

Expected Output

The command should complete successfully and display that the cache has been cleaned.

Figure 2.6: Cleaning YUM Cache



Step 7: Create Repository Metadata

Explanation

The **createrepo** command generates the repository metadata required by YUM. Without this metadata, YUM cannot identify the available software packages.

Command

createrepo /repo

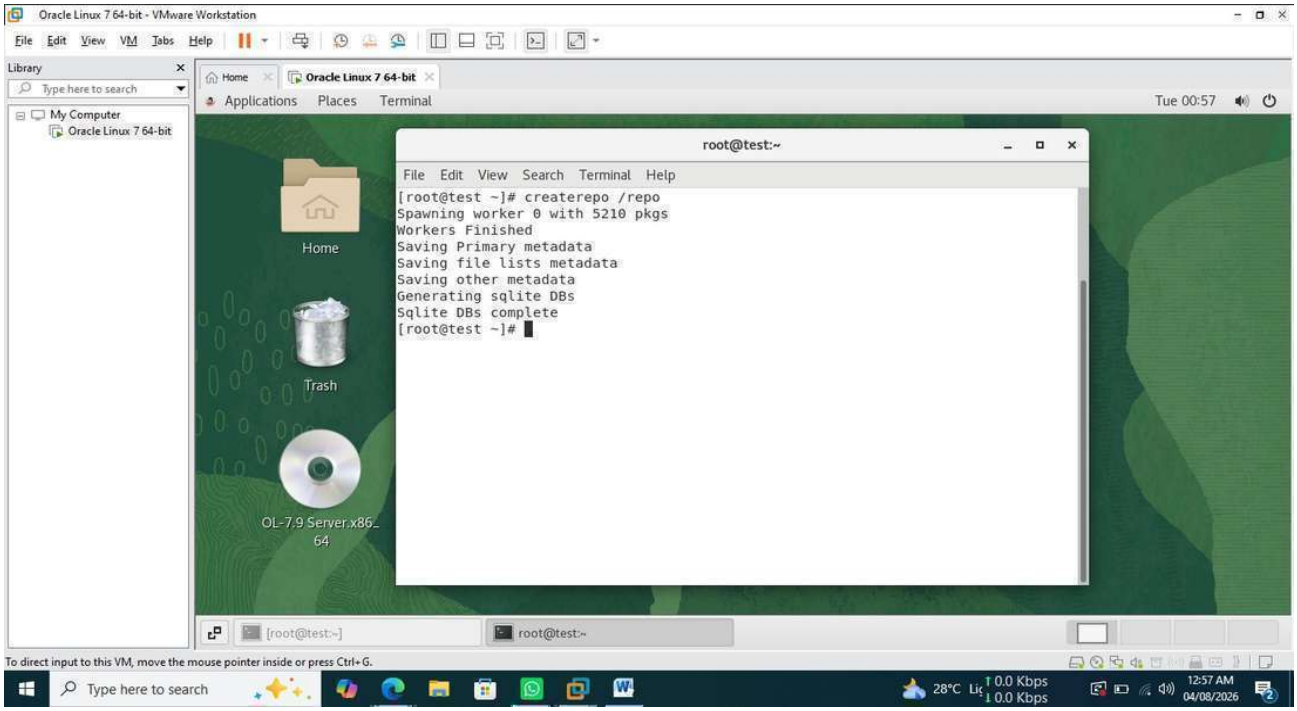
Command Explanation

- createrepo → Creates repository metadata.
- /repo → Directory containing the copied Oracle Linux packages.

Expected Output

The command should complete successfully and display messages indicating that repository metadata has been created.

Figure 2.7: Creating Repository Metadata



Step 8: Refresh the Repository Cache

Explanation

After creating the repository metadata, refresh the repository cache so that YUM recognizes the new local repository.

Commands

yum makecache

yum repolist

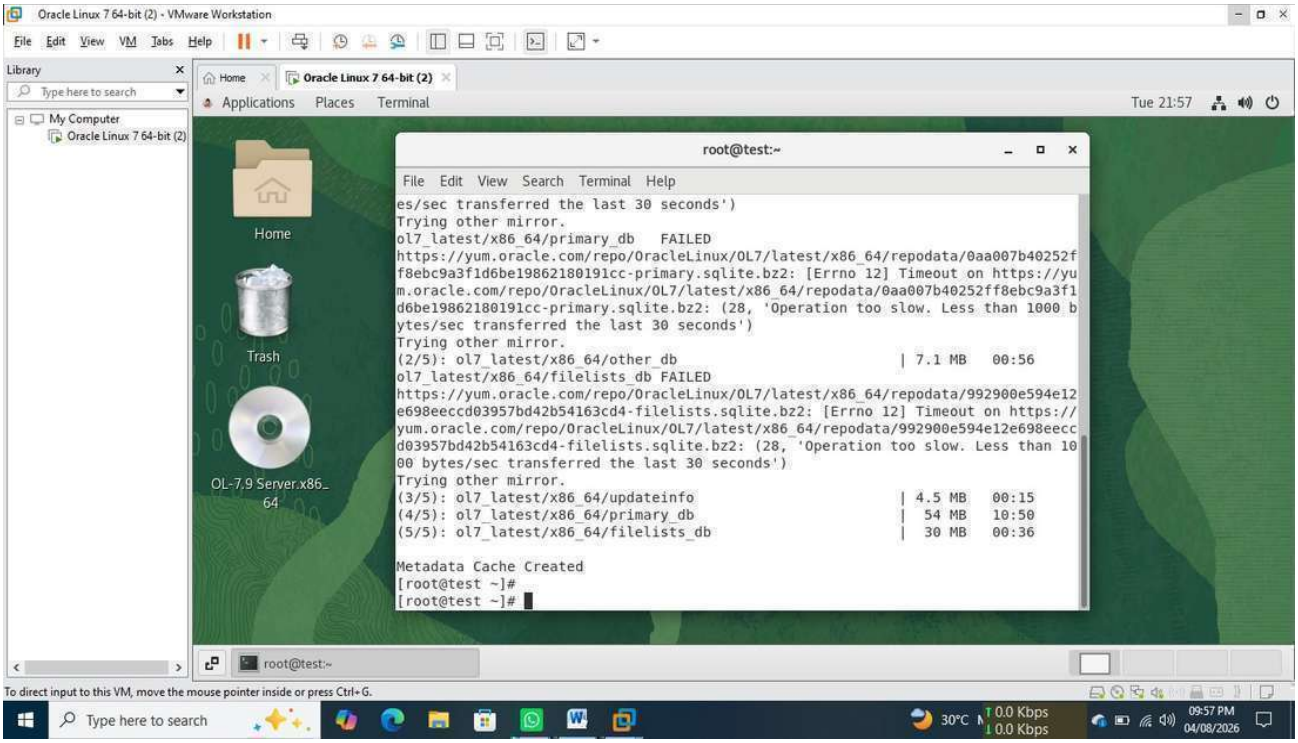
Command Explanation

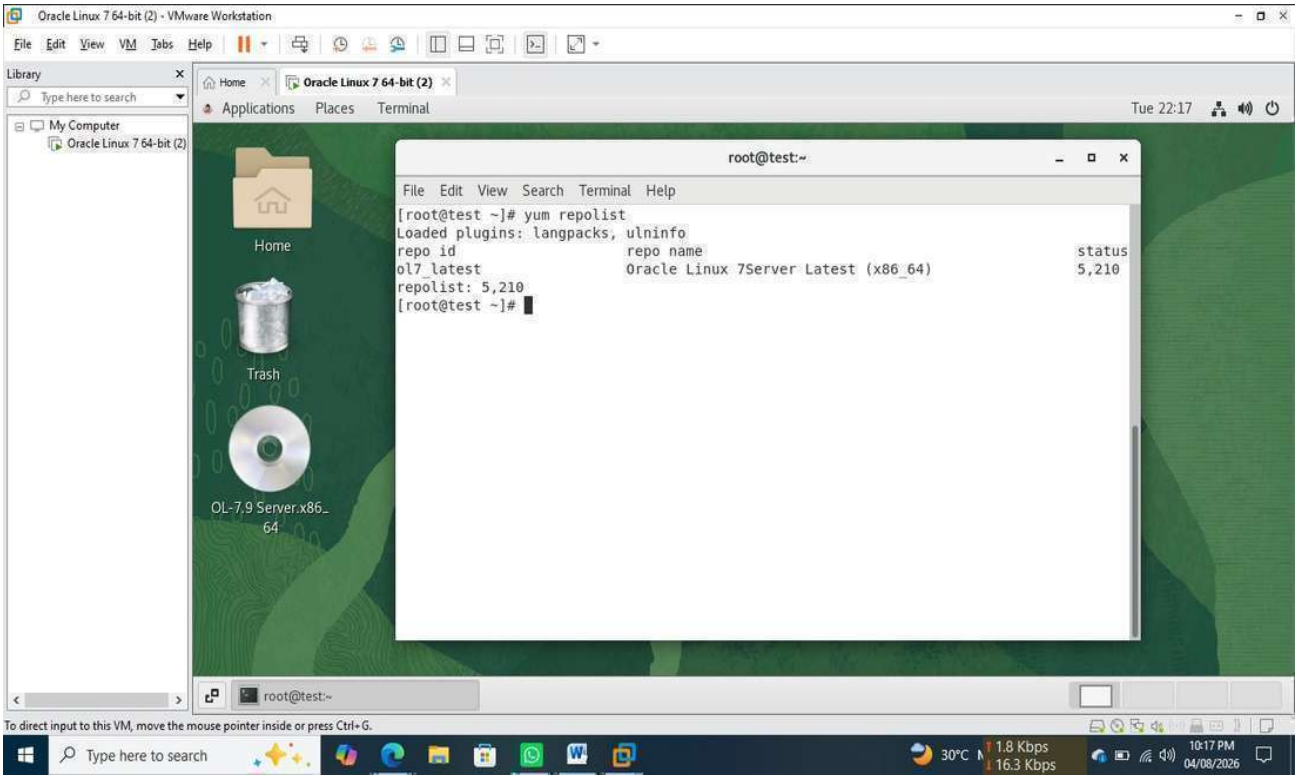
- yum makecache → Creates a fresh package cache.
- yum repolist → Displays the list of enabled repositories.

Expected Output

The configured local repository should appear in the repository list.

Figure 2.8: Refreshing the Local YUM Repository





2.4 Installing Oracle Pre-installation Packages

Introduction

After successfully configuring the Local YUM Repository, the next step is to install the Oracle pre-installation packages. These packages automatically install the required software dependencies, create the **oracle** user, create the **oinstall** and **dba** groups, and configure the operating system for Oracle Database installation.

This method is recommended because it reduces manual configuration and ensures that the system is properly prepared before installing Oracle Database 12c.

Step 9: Install Oracle Pre-installation Packages

Explanation

Use the YUM package manager to install all Oracle prerequisite packages. During this process, Oracle Linux automatically installs the required RPM packages and creates the Oracle installation user and groups.

Command

yum install -y oracle*

Command Explanation

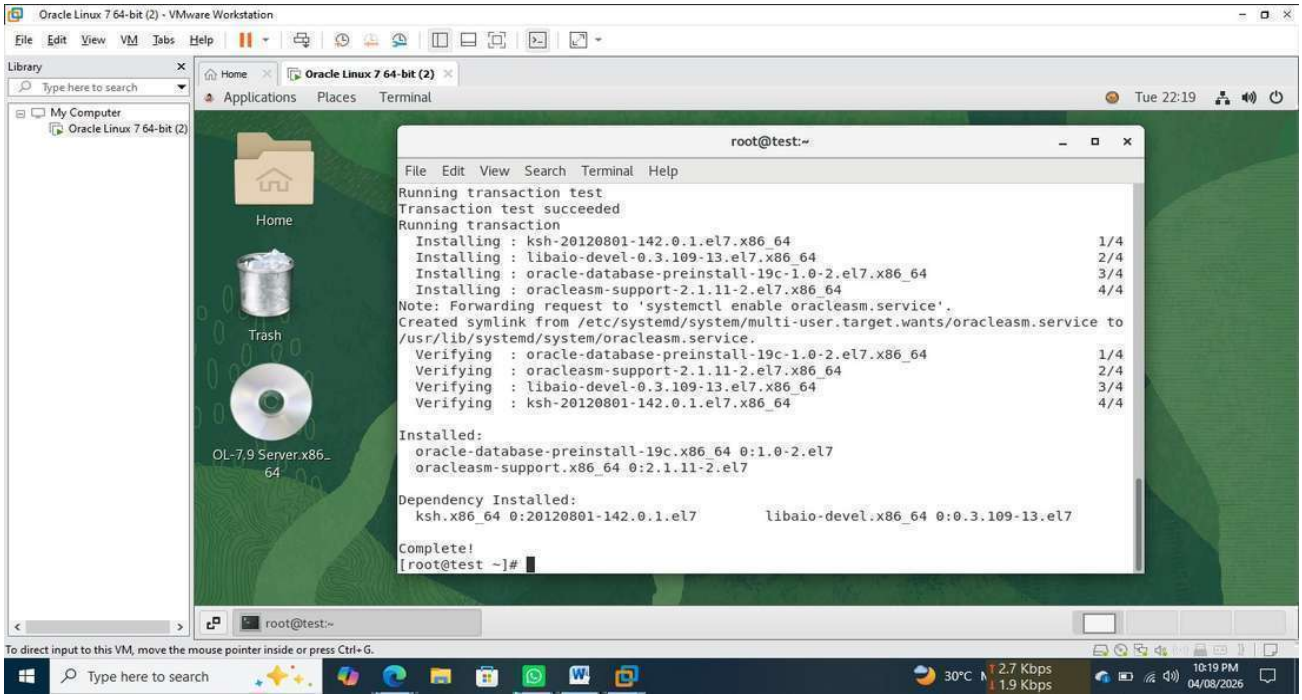
- **yum** → Package manager used in Oracle Linux.
- **install** → Installs the specified packages.
- **-y** → Automatically answers "Yes" to all prompts.
- **oracle*** → Installs all Oracle-related packages available in the configured repository.

Expected Output

The installation process should complete successfully and display:

Complete!

Figure 2.9: Installing Oracle Pre-installation Packages



Step 10: Verify Oracle User and Required Groups

Explanation

After the installation is complete, verify that the **oracle** user and the required groups (**oinstall** and **dba**) have been created successfully.

Command 1

id oracle

Expected Output

The output should display the Oracle user along with the **oinstall** and **dba** groups.

Command 2

grep oracle /etc/passwd

Expected Output

The Oracle user's account information should be displayed.

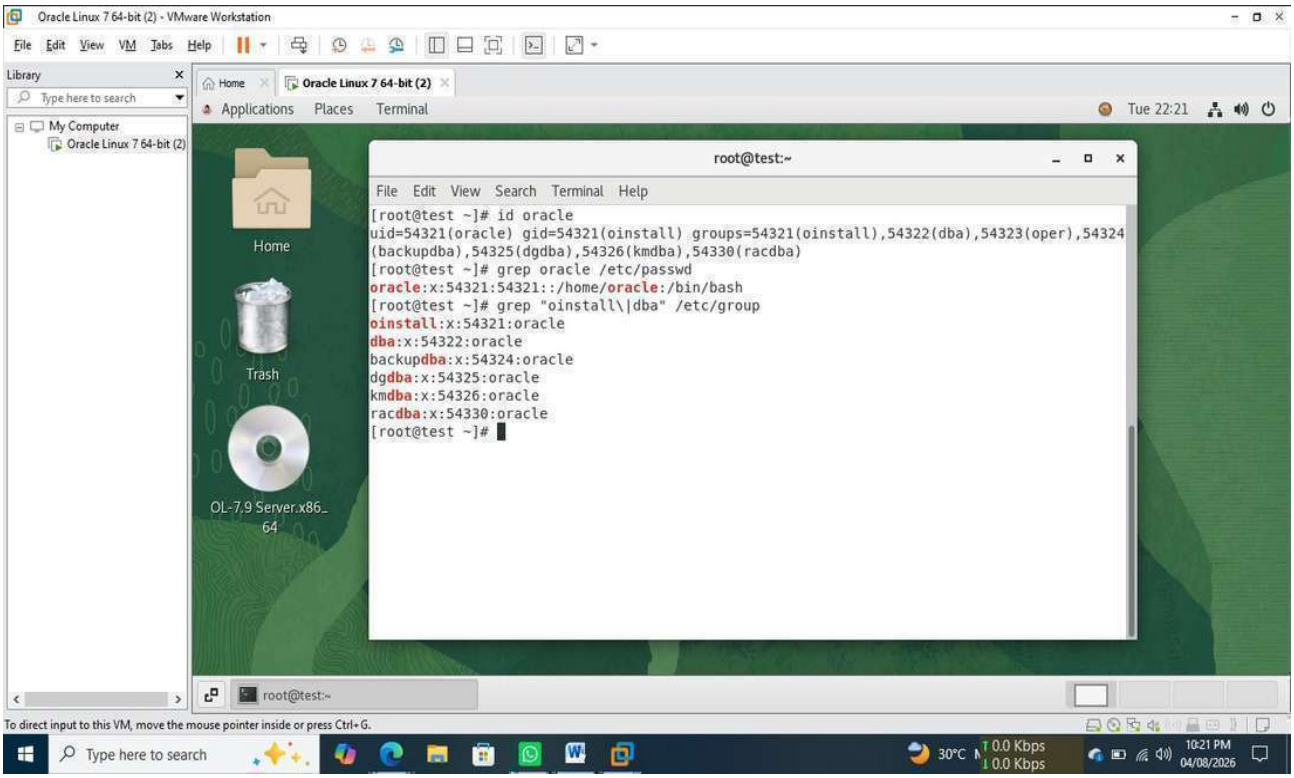
Command 3

grep "oinstall\|dba" /etc/group

Expected Output

The **oinstall** and **dba** groups should be listed.

Figure 2.10: Verifying Oracle User and Groups



Step 11: Disable SELinux

Explanation

SELinux (Security-Enhanced Linux) provides additional security by enforcing access control policies. However, during Oracle Database installation, it may block certain installation activities. Therefore, SELinux is temporarily disabled before proceeding with the installation.

Command

vi /etc/selinux/config

Locate the following line:

SELINUX=enforcing

Replace it with:

SELINUX=disabled

Save the file and exit.

Verification Command

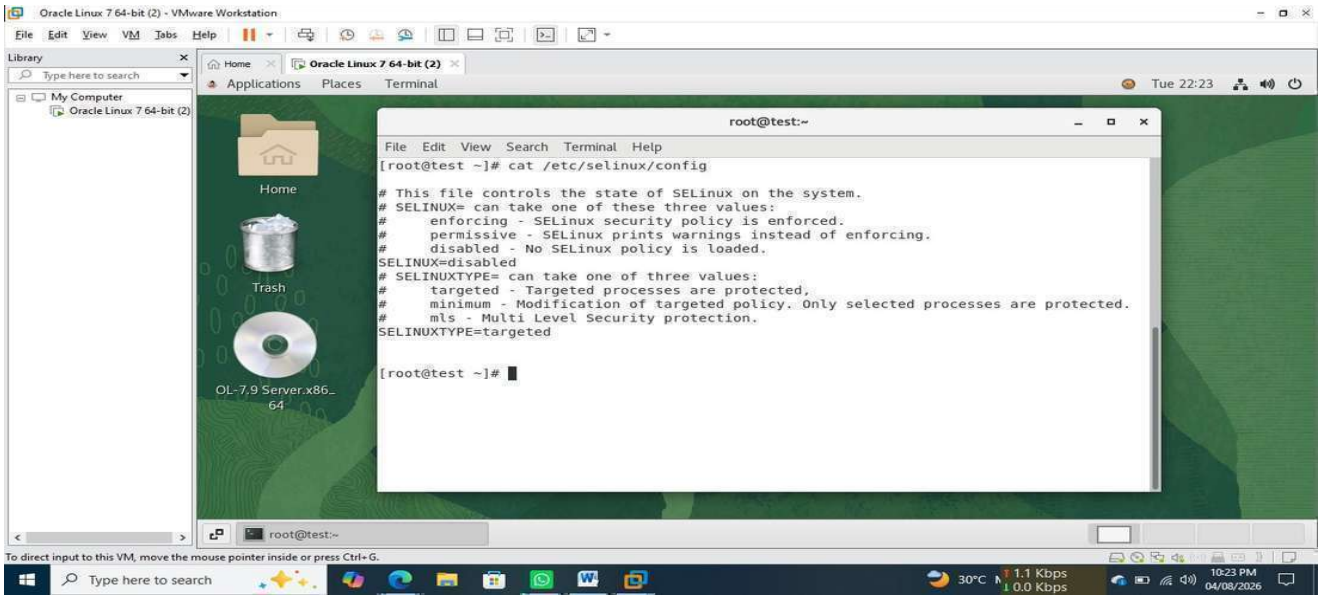
cat /etc/selinux/config

Expected Output

The configuration file should contain:

SELINUX=disabled

Figure 2.11: Disabling SELinux



Step 12: Restart Oracle Linux

Explanation

Restart the operating system so that the SELinux configuration changes take effect.

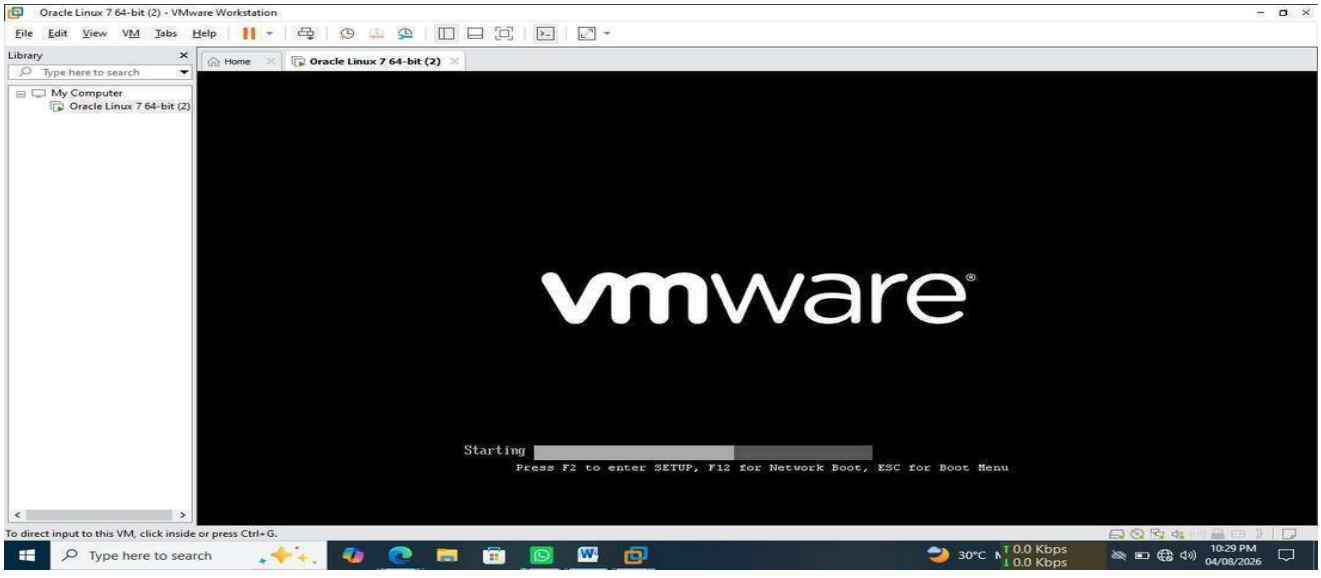
Command

reboot

Expected Result

The virtual machine should restart successfully. After rebooting, log in again using the **root** user to continue the installation process.

Figure 2.12: Oracle Linux Restart



2.5 Preparing Oracle Installation Environment

Introduction

After installing the Oracle prerequisite packages and restarting the system, the next step is to prepare the Oracle installation environment. This includes creating the required directories, copying the Oracle Database installation files, extracting the ZIP files, assigning the correct ownership and permissions, and preparing the Oracle user for software installation.

Step 13: Create Oracle Installation Directories

Explanation

Create the required directories that will be used during the Oracle Database installation. The **/backup** directory stores the Oracle installation files, while **/u01** is used as the Oracle Base directory.

Commands

```
mkdir /backup
mkdir /u01
```

Command Explanation

- **mkdir /backup** → Creates a directory to store Oracle installation files.
- **mkdir /u01** → Creates the Oracle Base directory.

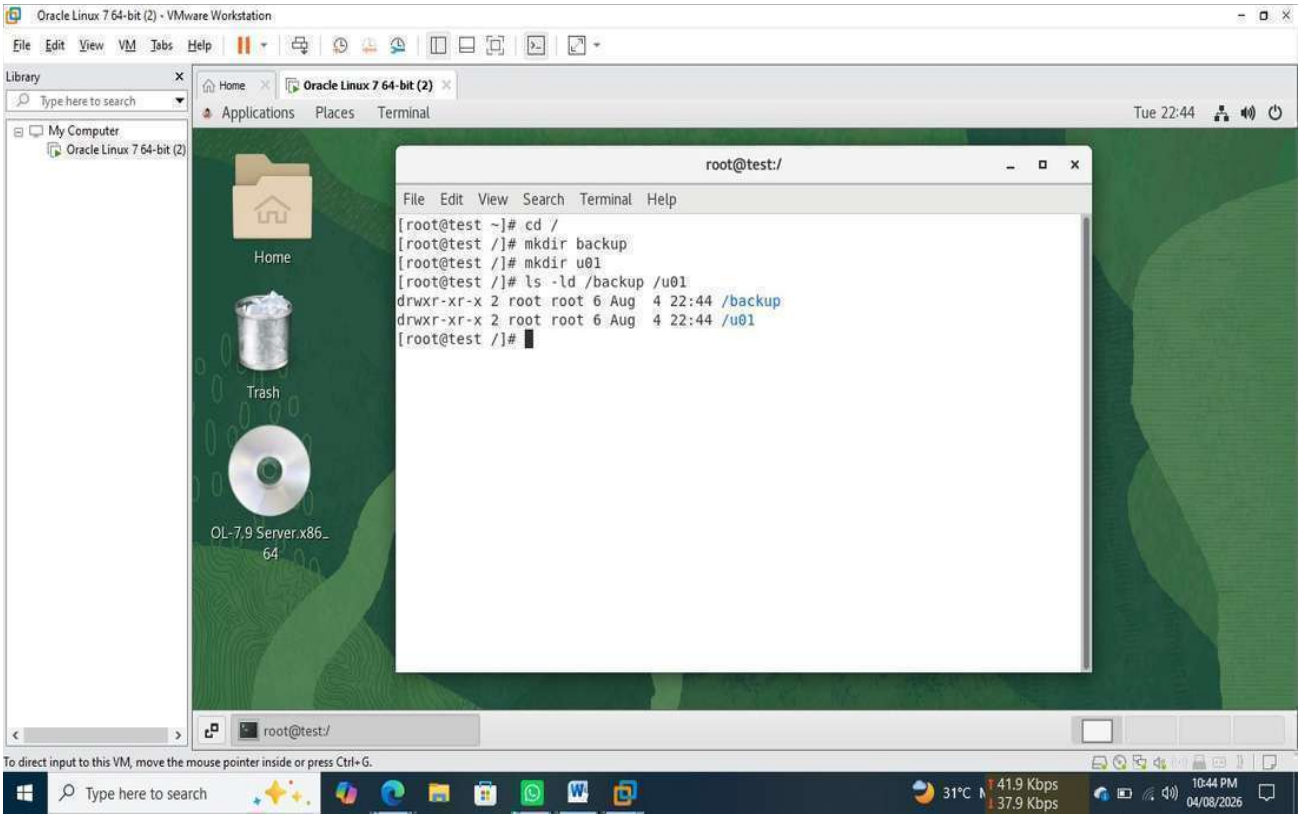
Verification Command

```
ls -ld /backup /u01
```

Expected Output

Both directories should be listed.

Figure 2.13: Creating Oracle Installation Directories



Step 14: Copy Oracle Database Installation Files

Explanation

Open the shared folder (**DB for Linux**) and copy both Oracle Database 12c ZIP files into the **/backup** directory.

Note: In our lab, the files were copied using the graphical file manager.

Explanation

cp -rvf "/mnt/hgfs/db for linux/"*.zip /backup/

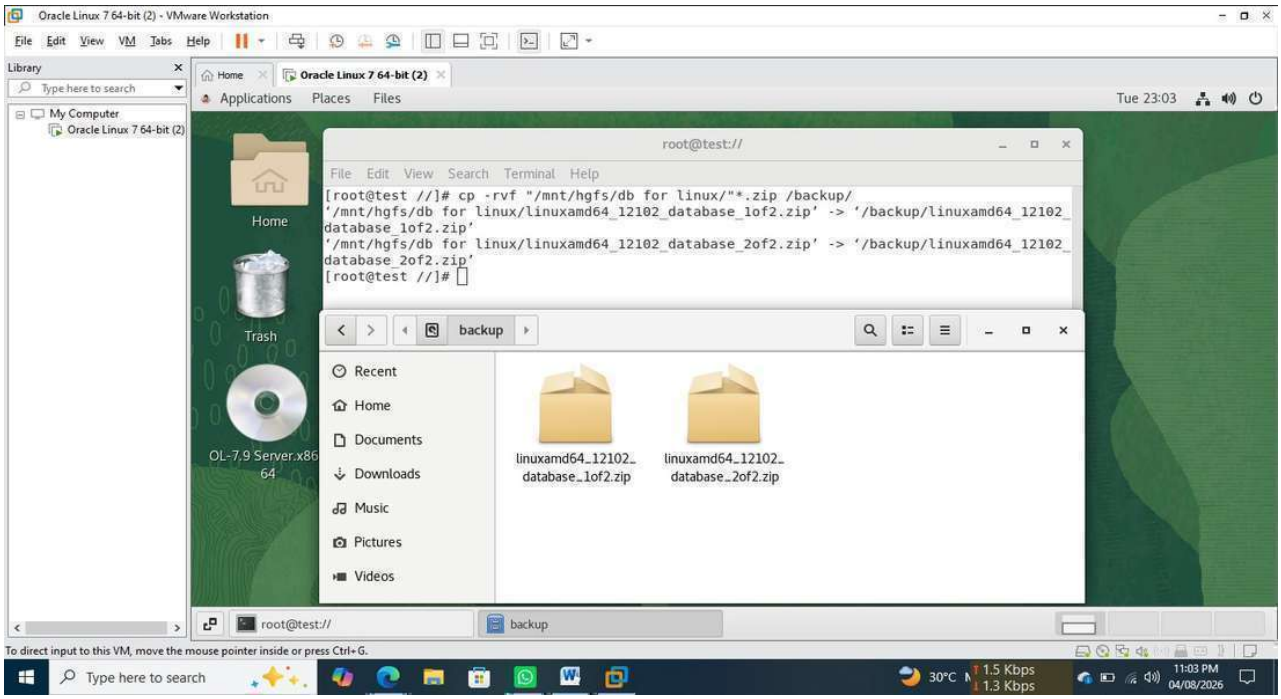
Command Explanation

- cp → Files copy karne ke liye.
- -r → Recursive copy (agar folders bhi hon).
- -v → Har file ka naam dikhata hai jab copy hoti hai.
- -f → Existing file ho to overwrite kar deta hai.
- "/mnt/hgfs/db for linux/"*.zip → Shared folder me maujood sab .zip files.
- /backup/ → Destination folder.

Verification

Open the **/backup** directory and verify that both ZIP files are present.

Figure 2.14: Copying Oracle Database Installation Files



Step 15: Extract Oracle Database ZIP Files

Explanation

Extract both Oracle Database ZIP files. After extraction, a folder named **database** will be created automatically.

Commands

```
cd /backup
unzip linuxamd64_12102_database_1of2.zip
unzip linuxamd64_12102_database_2of2.zip
```

Note: If your ZIP files have different names, use the actual filenames.

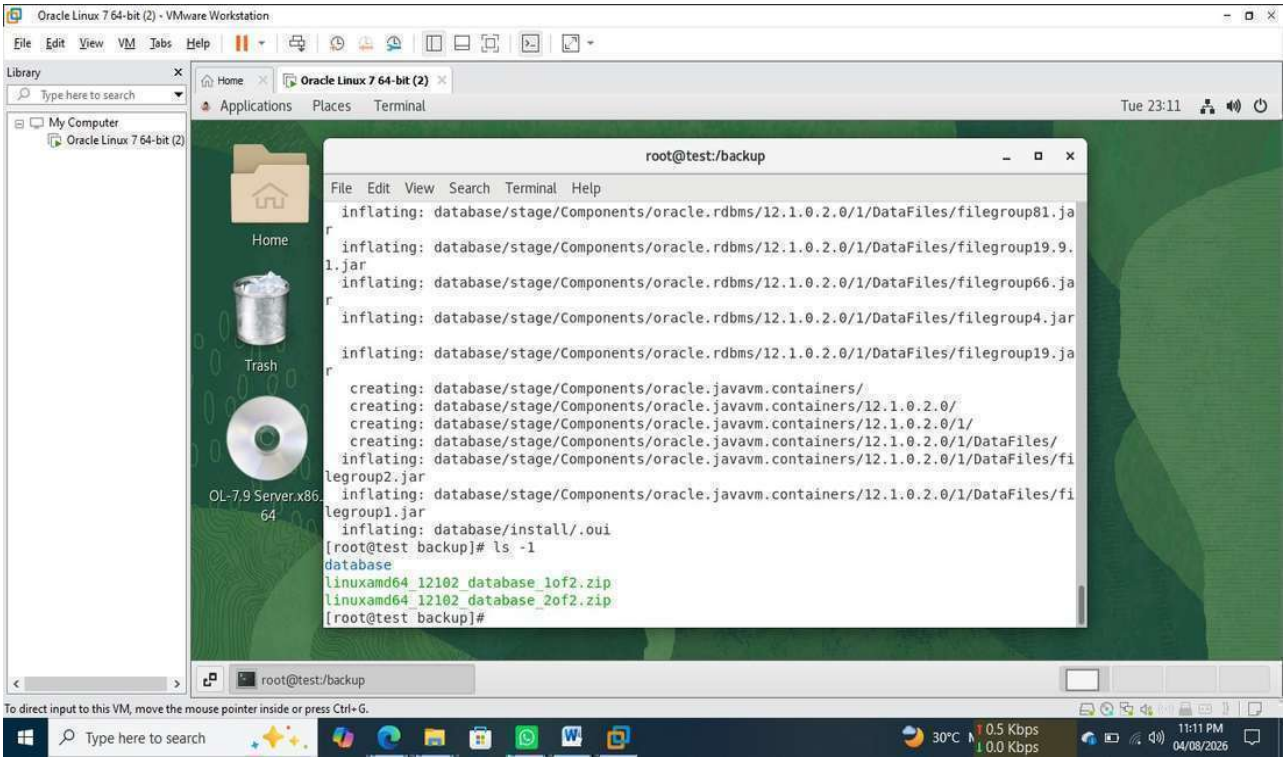
Verification Command

ls -l

Expected Output

The **database** directory should appear.

Figure 2.15: Extracting Oracle Database Installation Files



Step 16: Remove the Temporary Test User

Explanation

The **test** user was created during the Oracle Linux installation for initial system setup. Since Oracle Database administration will now be performed using the **oracle** user, the temporary **test** user is removed.

Command

userdel -r test

Command Explanation

- **userdel** → Deletes a user account.
- **-r** → Removes the user's home directory and mail spool.

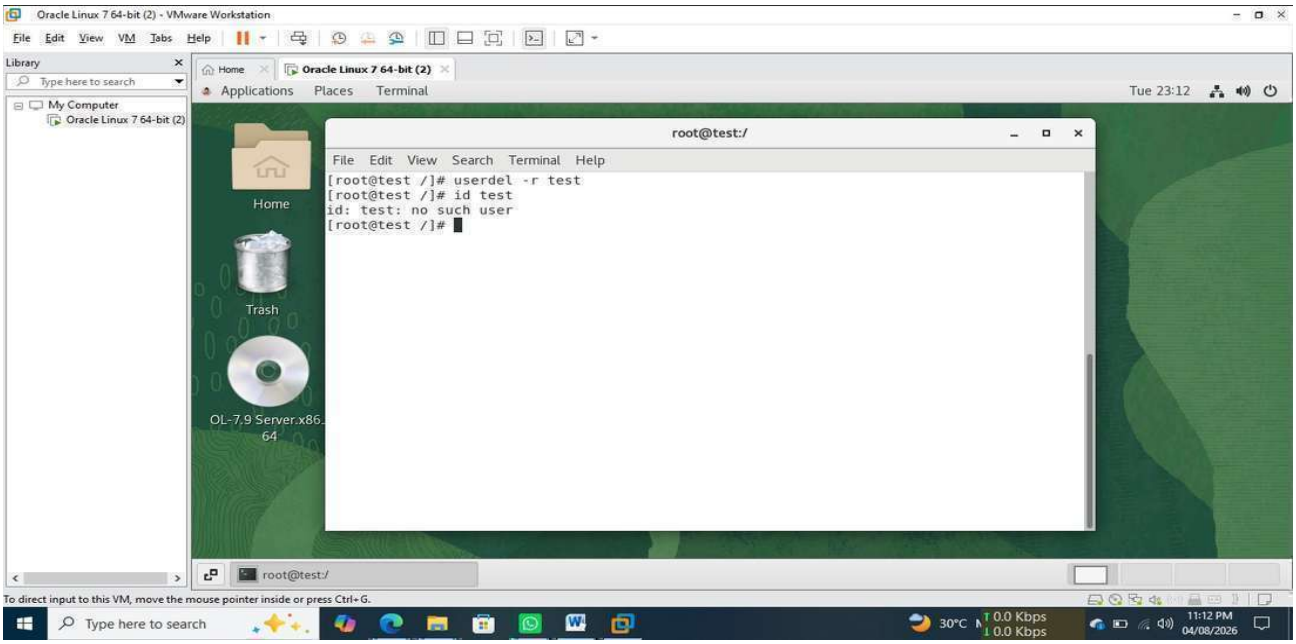
Verification Command

id test

Expected Output

id: test: no such user

Figure 2.16: Removing the Test User



Step 17: Configure Directory Ownership

Explanation

Assign ownership of the Oracle installation directories to the **oracle** user and the **oinstall** group.

Commands

chown -R oracle:oinstall /backup

chown -R oracle:oinstall /u01

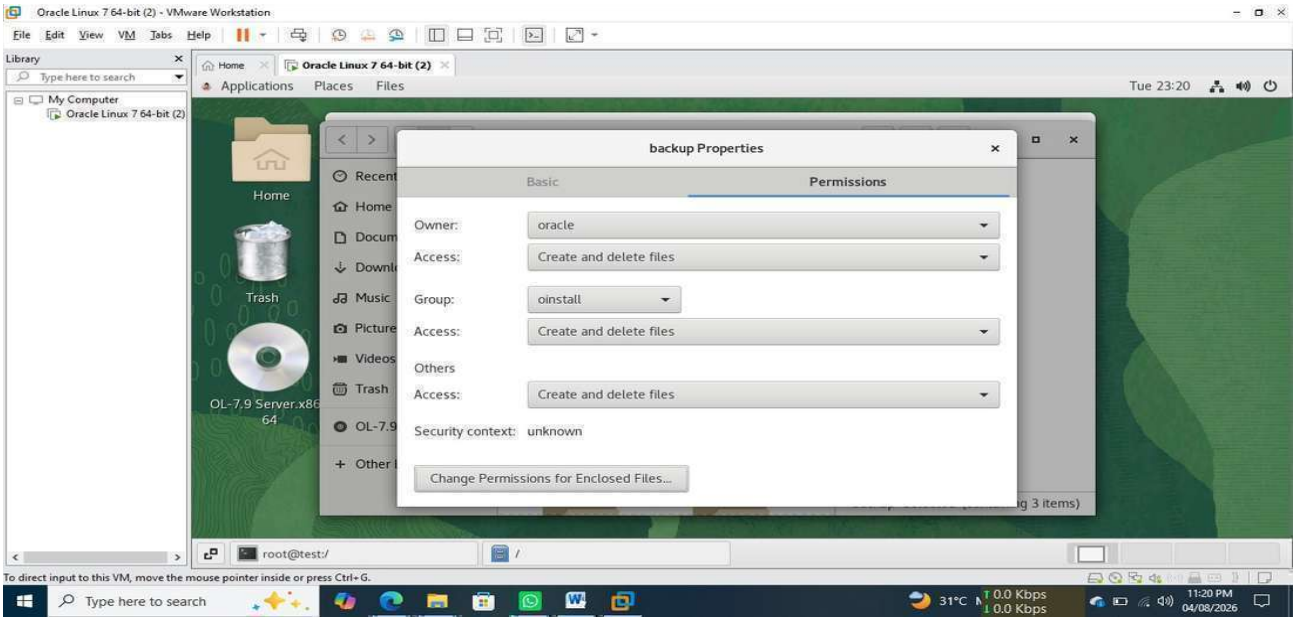
Verification Command

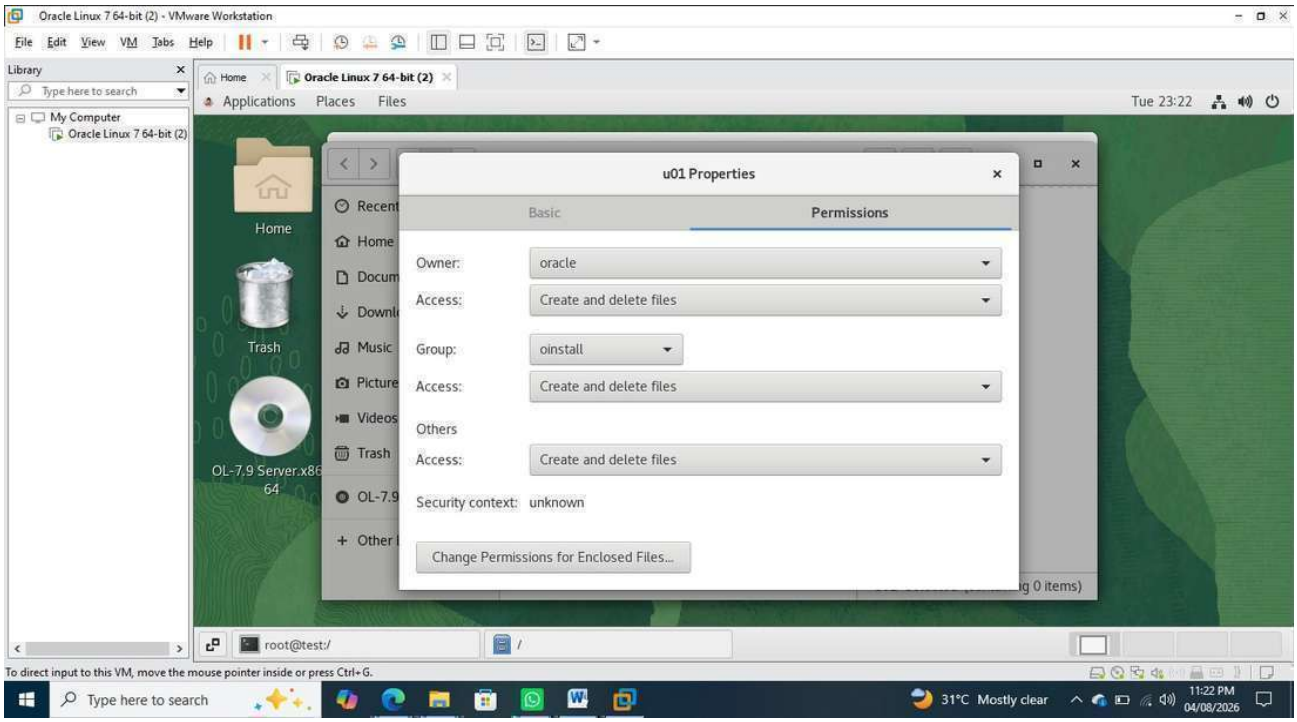
ls -ld /backup /u01

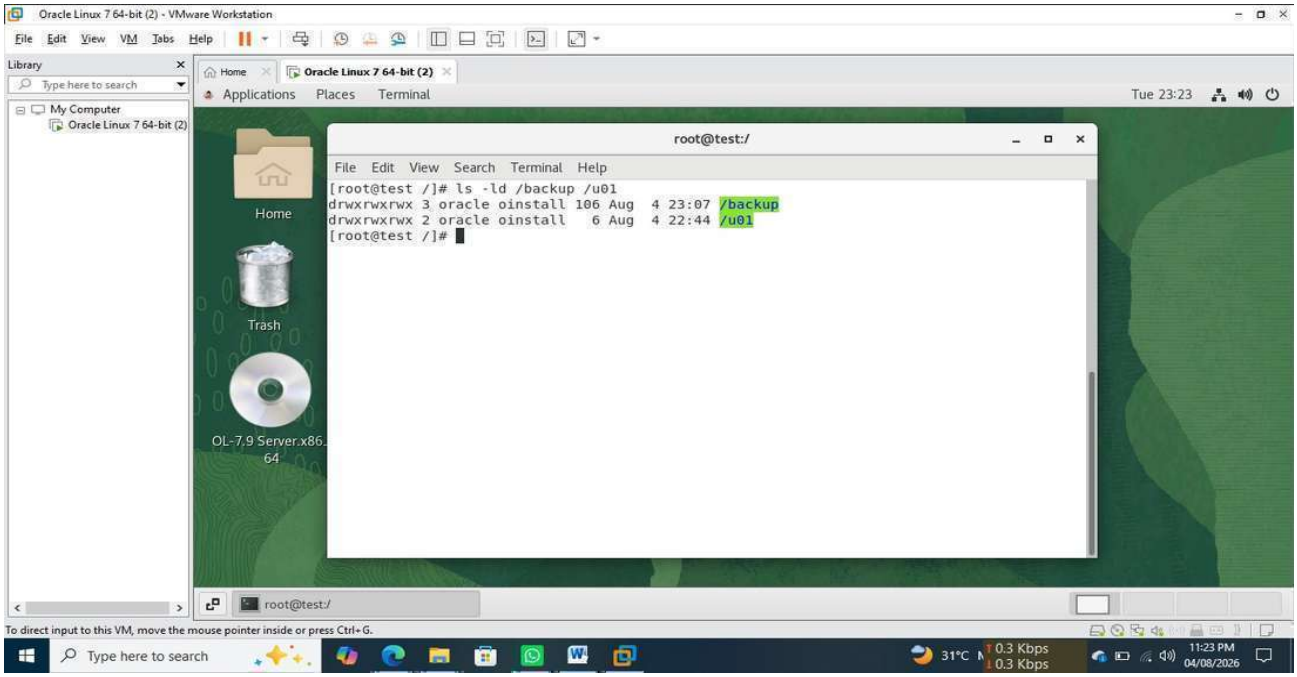
Expected Output

The owner should be **oracle** and the group should be **oinstall**.

Figure 2.17: Configuring Directory Ownership







Step 18: Configure Directory Permissions

Explanation

Grant the required permissions so that the Oracle user can access and manage the installation directories.

Commands

chmod -R 775 /backup

chmod -R 775 /u01

Command Explanation

- **chmod** → Changes file or directory permissions.
- **775** → Grants read, write and execute permissions to the owner and group, and read/execute permissions to others.

Verification Command

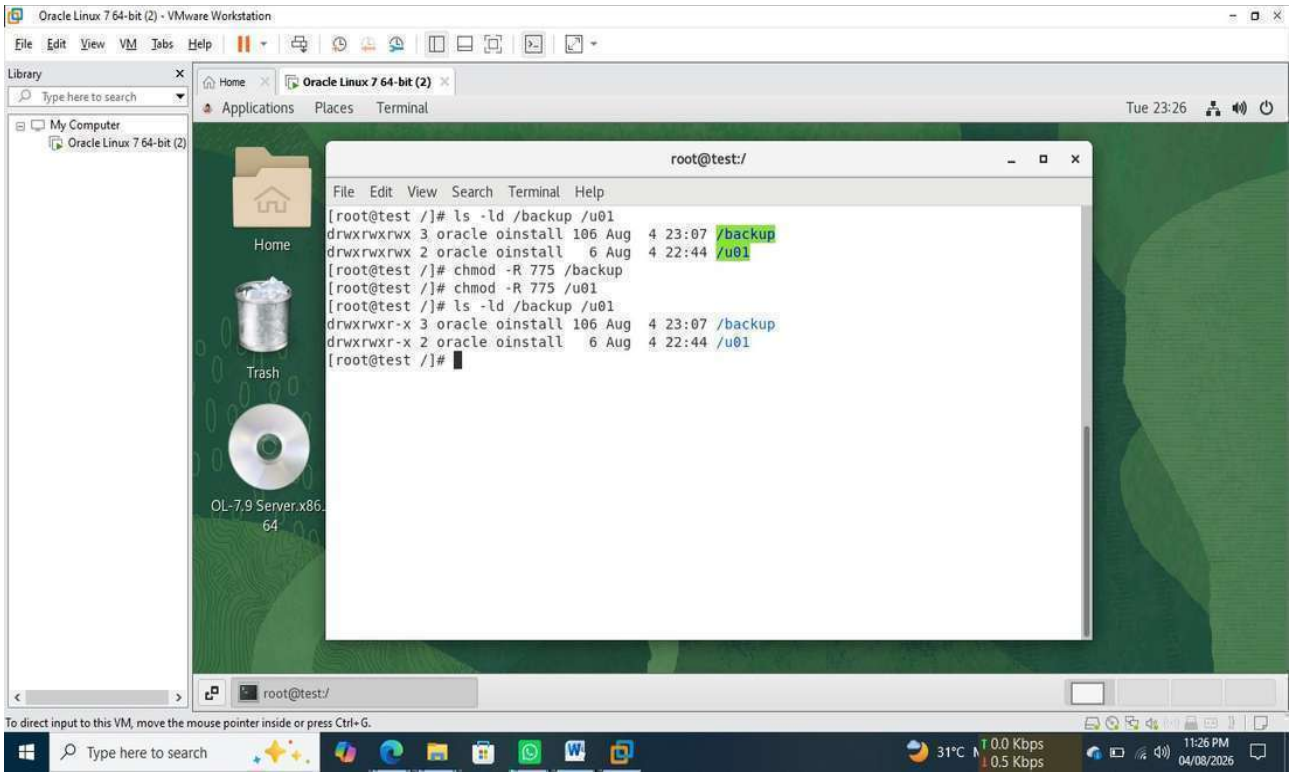
ls -ld /backup /u01

Expected Output

The permission should be displayed as:

drwxrwxr-x

Figure 2.18: Configuring Directory Permissions



2.6 Launching Oracle Universal Installer

Introduction

After preparing the Oracle installation environment, the final step is to launch the Oracle Universal Installer (OUI). The installer must be executed using the **oracle** user because Oracle software should never be installed using the **root** account.

Step 19: Log in as Oracle User

Explanation

Log out from the **root** account and log in using the **oracle** user. The Oracle software owner is responsible for installing and managing Oracle Database software.

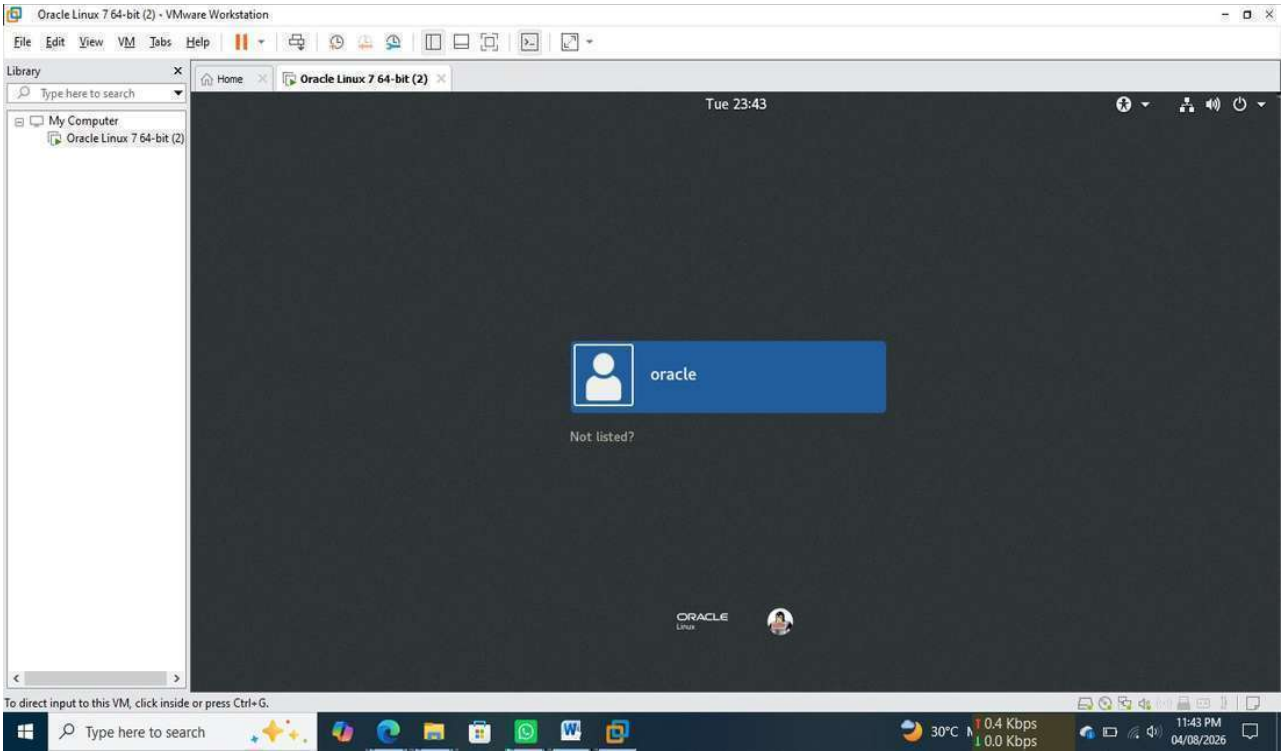
Verification Command

whoami

Expected Output

oracle

Figure 2.19: Login as Oracle User



Step 20: Navigate to the Oracle Database Installation Directory

Explanation

Move to the directory where the Oracle Database installation files were extracted.

Commands

cd /backup/database

Verification Command

pwd

Expected Output

/backup/database

Verify Installation Files

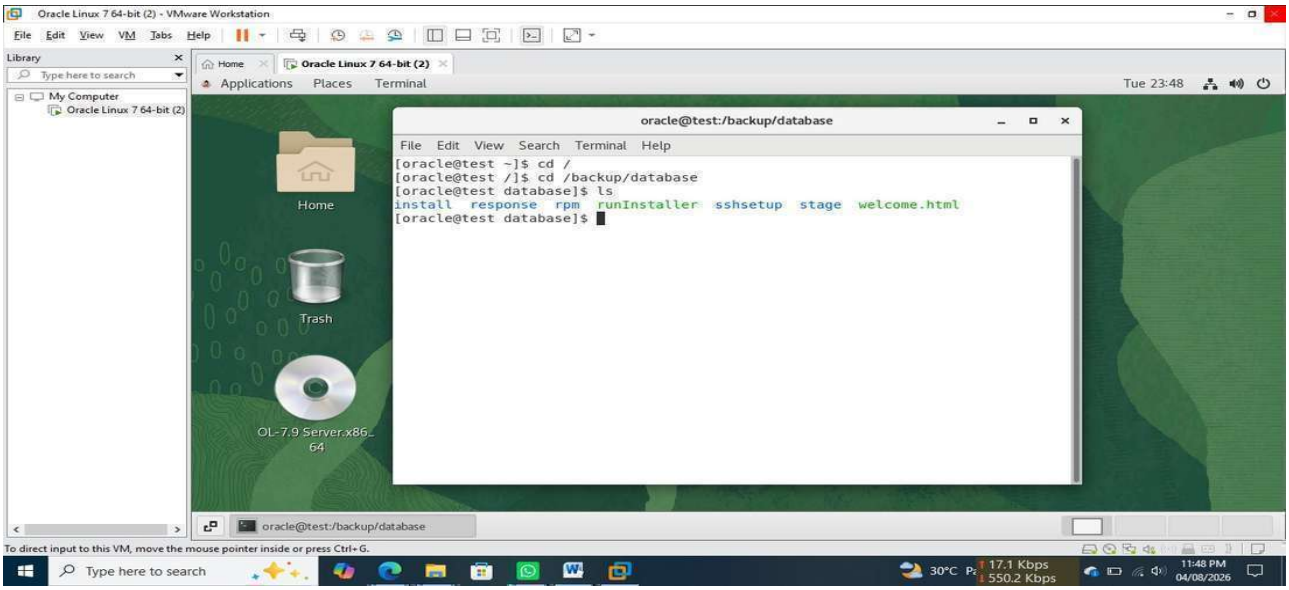
ls

Expected Output

The following files should be visible:

- runInstaller
- install
- response
- rpm
- stage
- sshsetup
- welcome.html

Figure 2.20: Oracle Database Installation Directory



Step 21: Launch Oracle Universal Installer

Explanation

Run the Oracle Universal Installer. This graphical installer will guide you through the Oracle Database 12c installation process.

Command

./runInstaller

Note: If the installer does not start because of a **DISPLAY** issue, configure the DISPLAY variable before running the installer.

Example:

export DISPLAY=:0

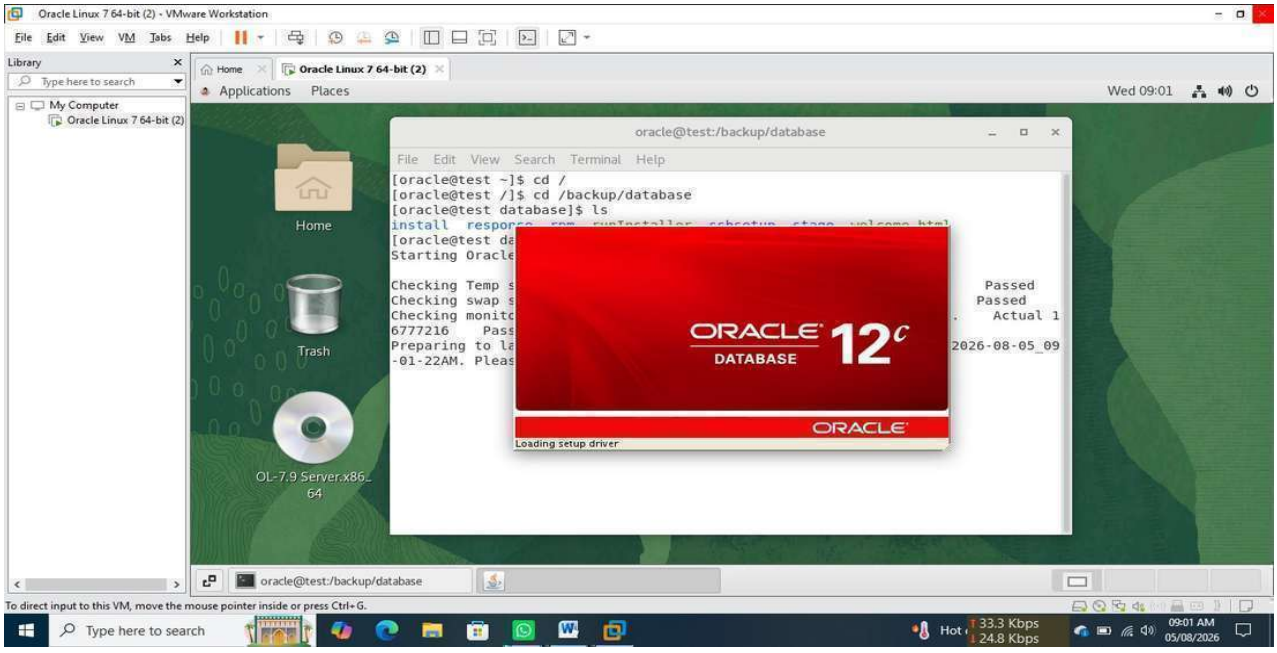
Then run:

./runInstaller

Expected Result

The Oracle Universal Installer window should open successfully.

Figure 2.21: Oracle Universal Installer



2.7 Configuring Oracle Universal Installer

Introduction

After launching the Oracle Universal Installer (OUI), a graphical installation wizard appears. This wizard guides the user through the Oracle Database 12c installation process. During this stage, different installation options are selected, Oracle directories are configured, prerequisite checks are performed, and the software installation begins.

The following steps describe each screen of the Oracle Universal Installer.

Step 22: Configure Security Updates

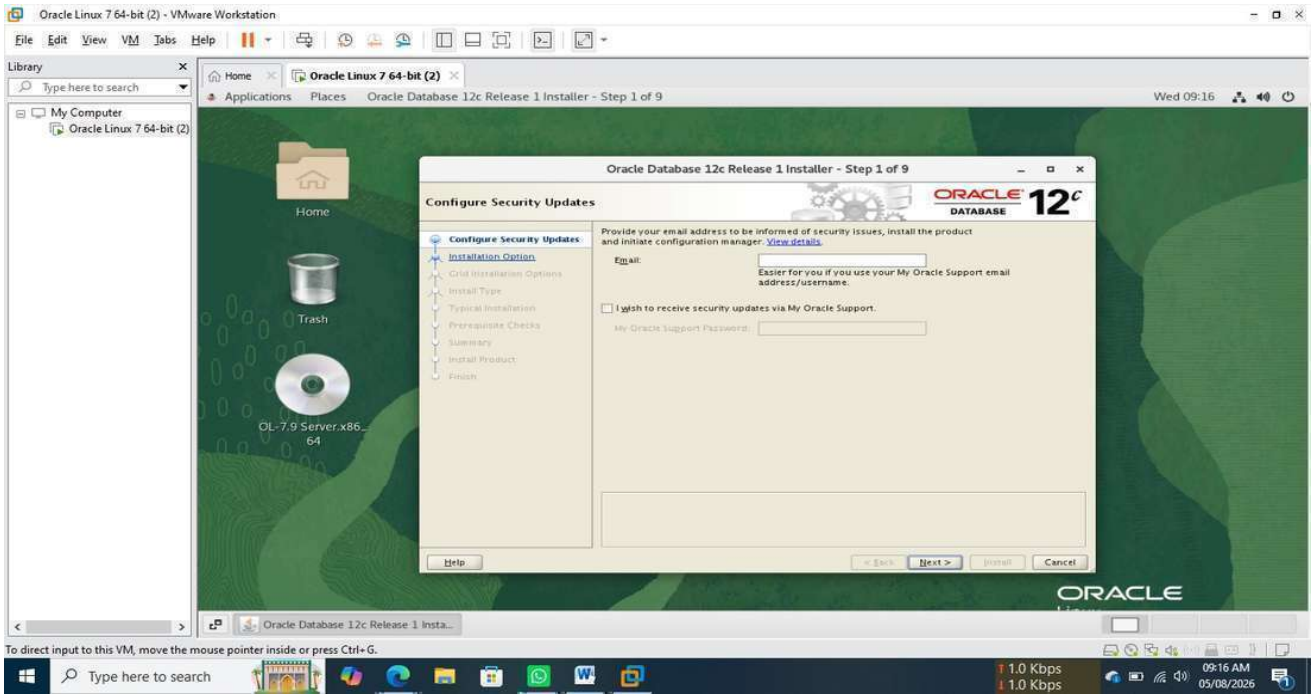
Explanation

The first screen of the Oracle Universal Installer asks for an email address to receive Oracle security updates. Since this installation is performed in a laboratory environment, the email notification option is not required.

Configuration

- Leave the Email field blank.
- Uncheck **"I wish to receive security updates via My Oracle Support."**
- Click **Next**.
- If a warning message appears, click **Yes** to continue.

Figure 2.22 : Select Installation Option



Step 23: Select Installation Option

Explanation

Choose the installation type according to the lab requirements.

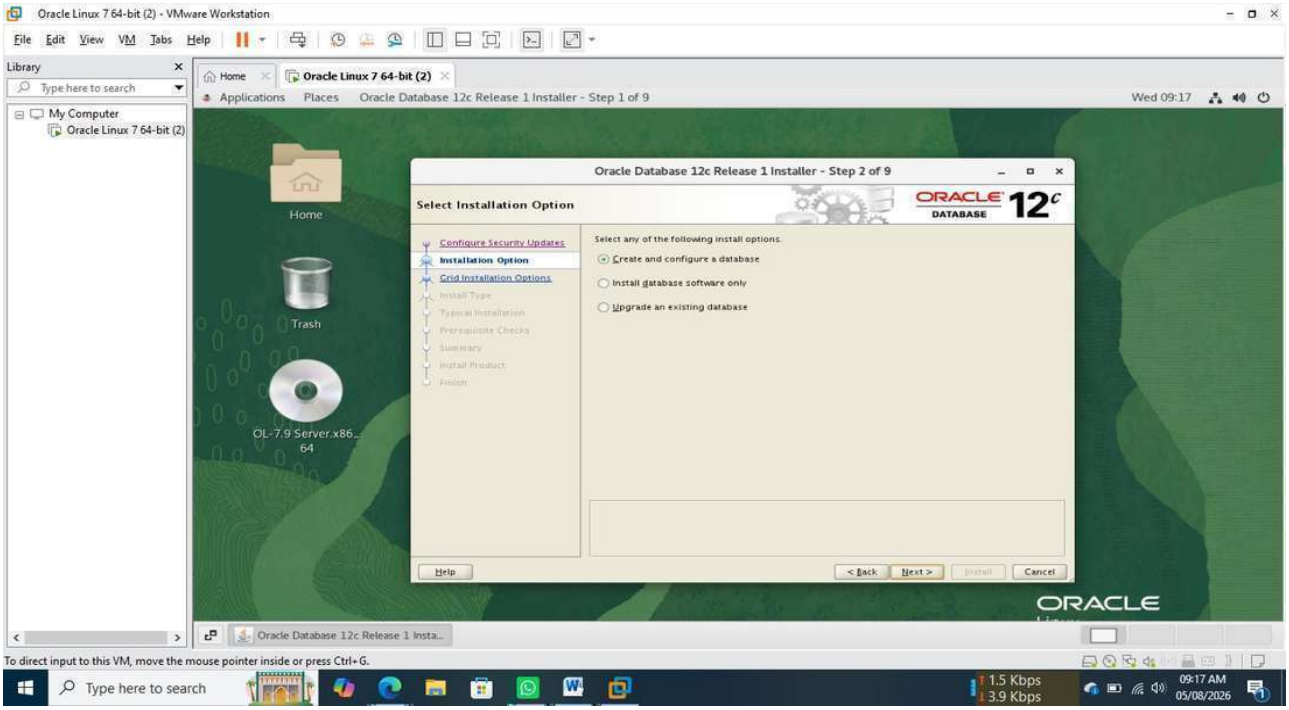
Configuration

Select:

- **Create and Configure a Database**

Click **Next**.

Figure 2.23: Select Installation Option



Step 24: Select System Class

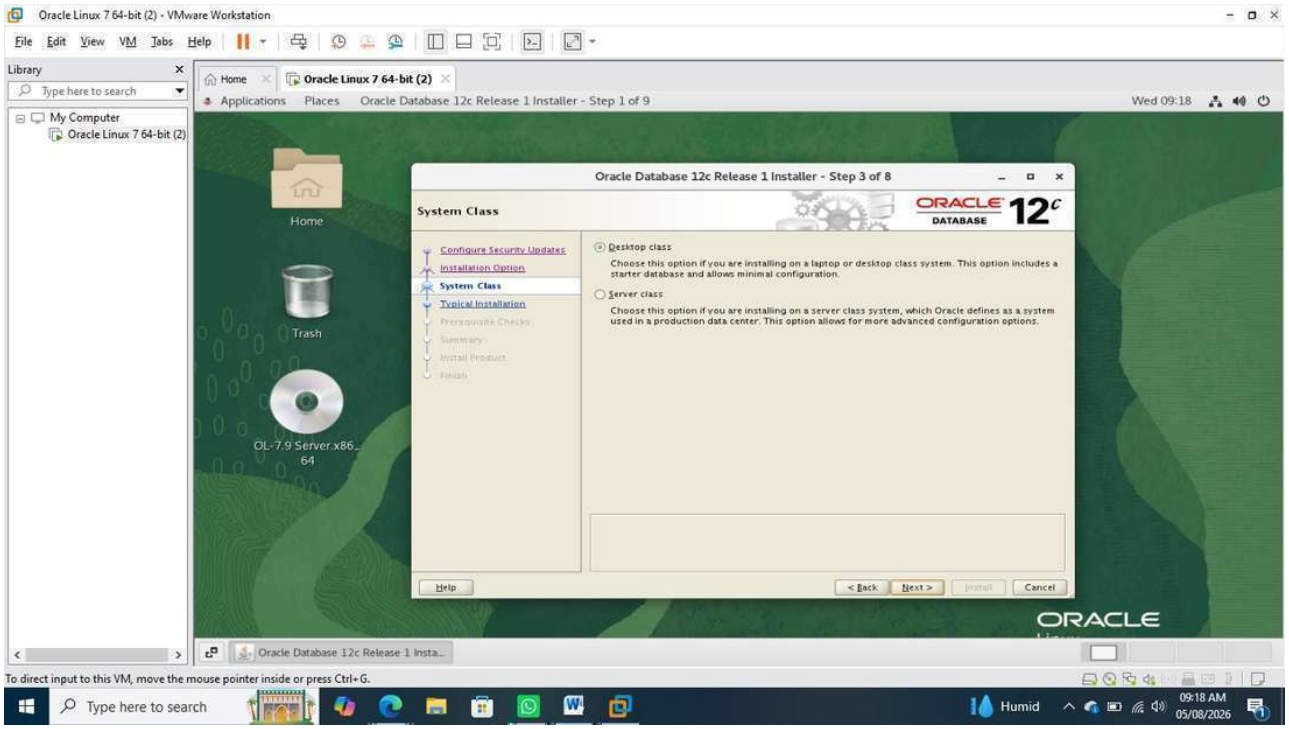
Explanation

Oracle Database can be installed for desktop or server environments. Since this practical is performed on a personal virtual machine, the Desktop Class option is selected.

Configuration

- Select **Desktop Class**.
- Click **Next**.

Figure 2.24: Select System Class



Step 25: Specify Oracle Home User

Explanation

The installer asks which operating system user will own the Oracle software.

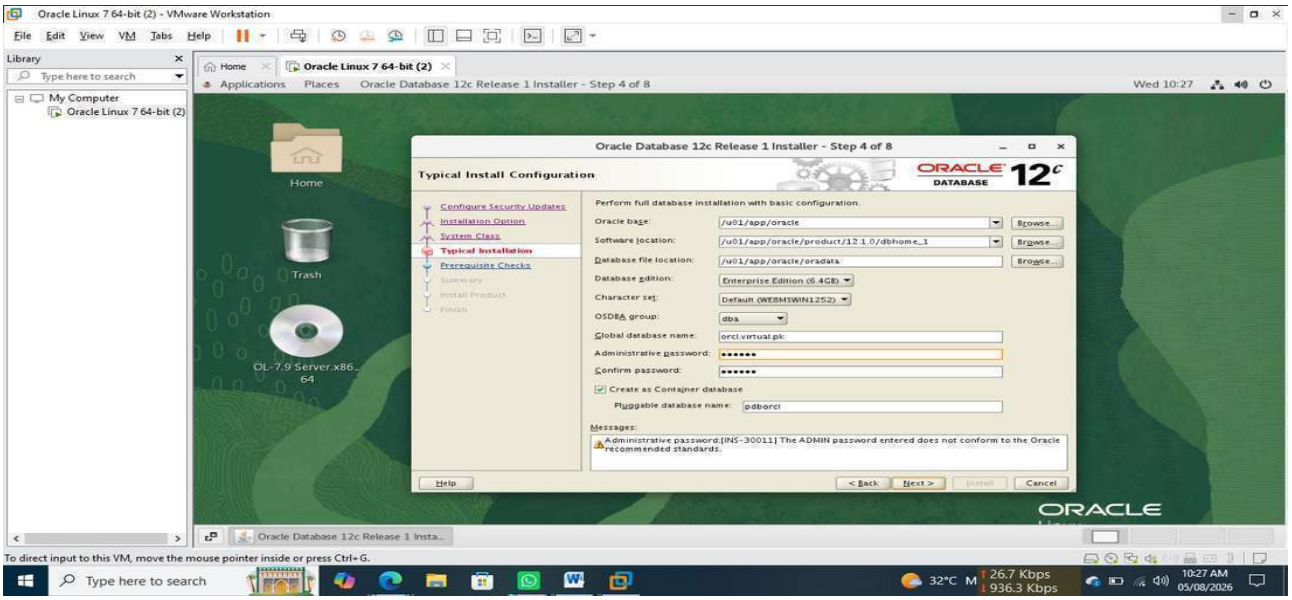
Configuration

Select:

- **Use Existing User**
- Username: **oracle**

Click **Next**.

Figure 2.25: Specify Oracle Home User



Step 26: Specify Installation Location

Explanation

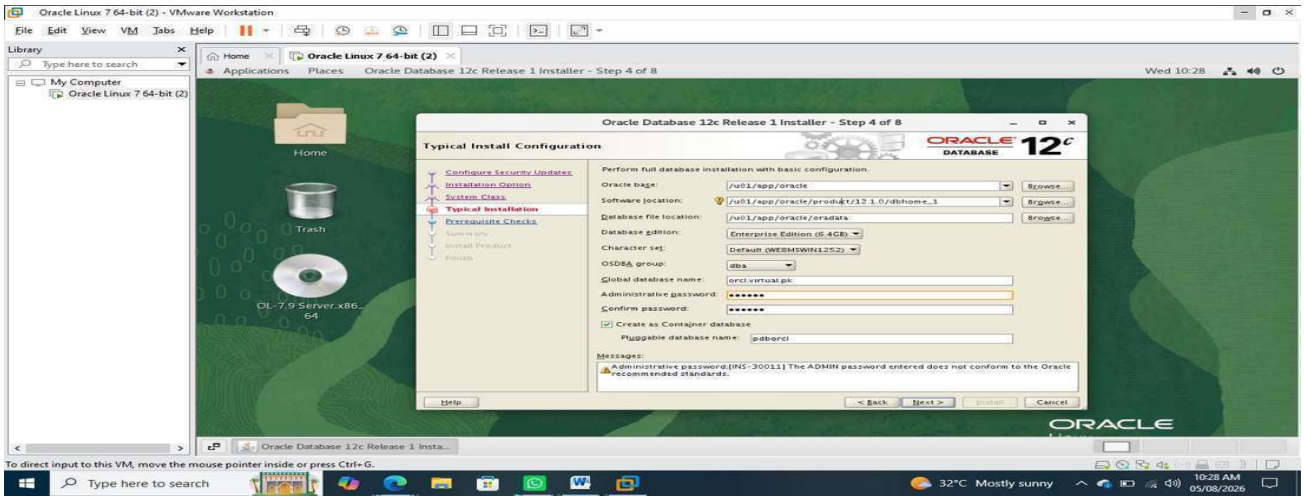
Provide the Oracle Base and Software Location where Oracle Database files will be installed.

Example Configuration

- **Oracle Base:** /u01/app/oracle
- **Software Location:** /u01/app/oracle/product/12.1.0/dbhome_1

Click **Next**.

Figure 2.26: Specify Installation Location



Step 27: Typical Installation Configuration

Explanation

In this step, Oracle Universal Installer asks for the database configuration details. These settings define the database name, administrative password, storage location, and character set. Enter the required information according to your lab requirements.

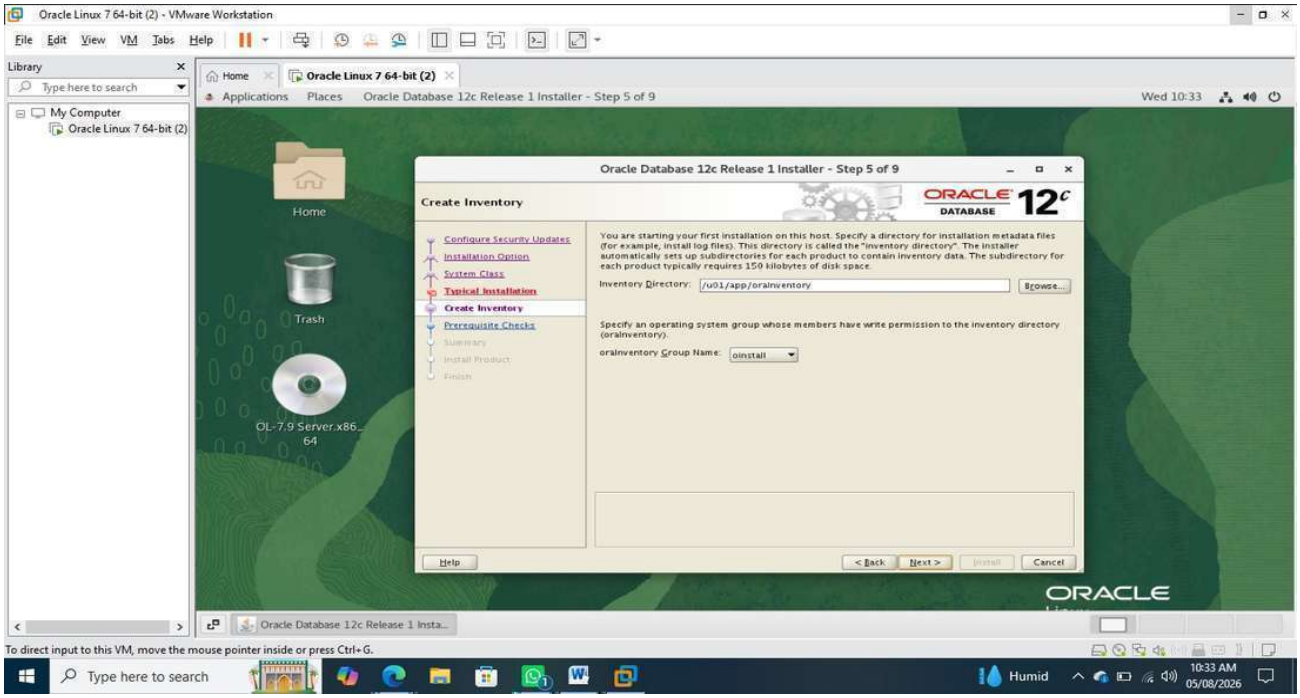
Example Configuration

Field	Value
Oracle Base	/u01/app/oracle
Database File Location	Default
Global Database Name	orcl
Administrative Password	Oracle
Confirm Password	oracle

Note: Use the password provided by your instructor. If a different password was used during the practical, enter that password instead.

Click **Next**.

Figure 2.27: Typical Installation Configuration



Step 28: Perform Prerequisite Checks

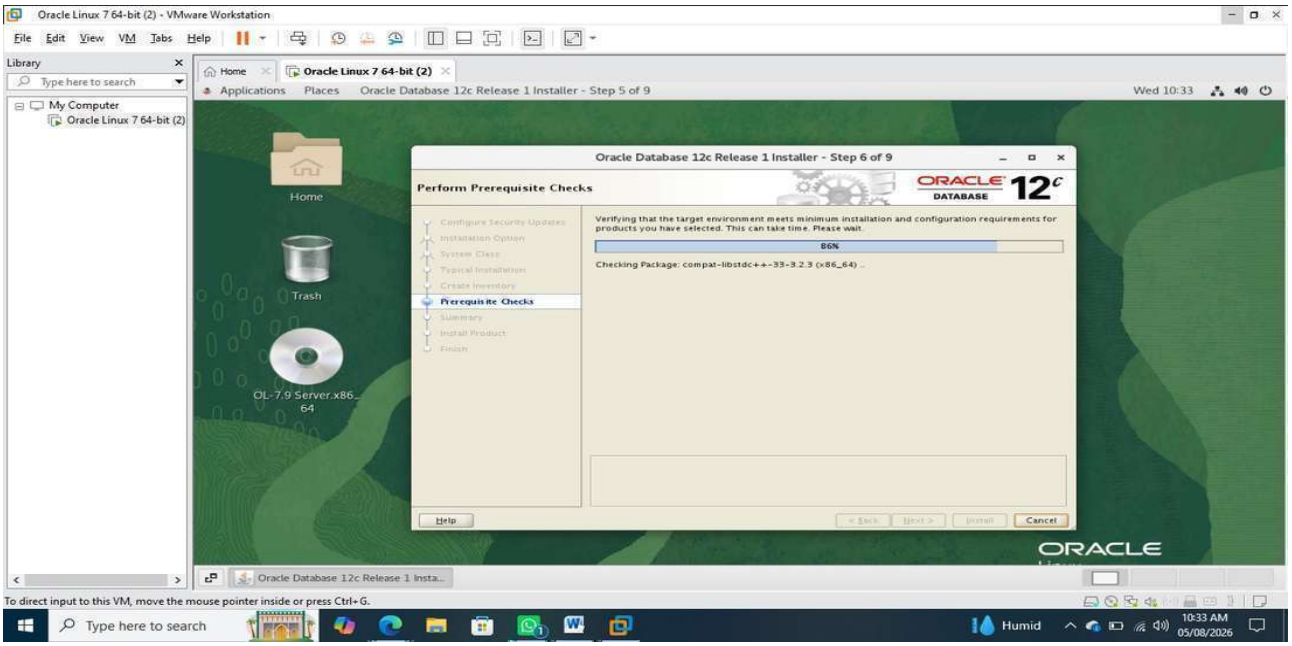
Explanation

The installer automatically checks whether the operating system meets all Oracle Database installation requirements. It verifies memory, disk space, packages, kernel parameters, swap space, and user permissions.

Expected Result

- If all checks pass successfully, click **Next**.
- If warning messages appear, review them and proceed according to your instructor's guidance.

Figure 2.28: Prerequisite Checks



Step 29: Review Summary

Explanation

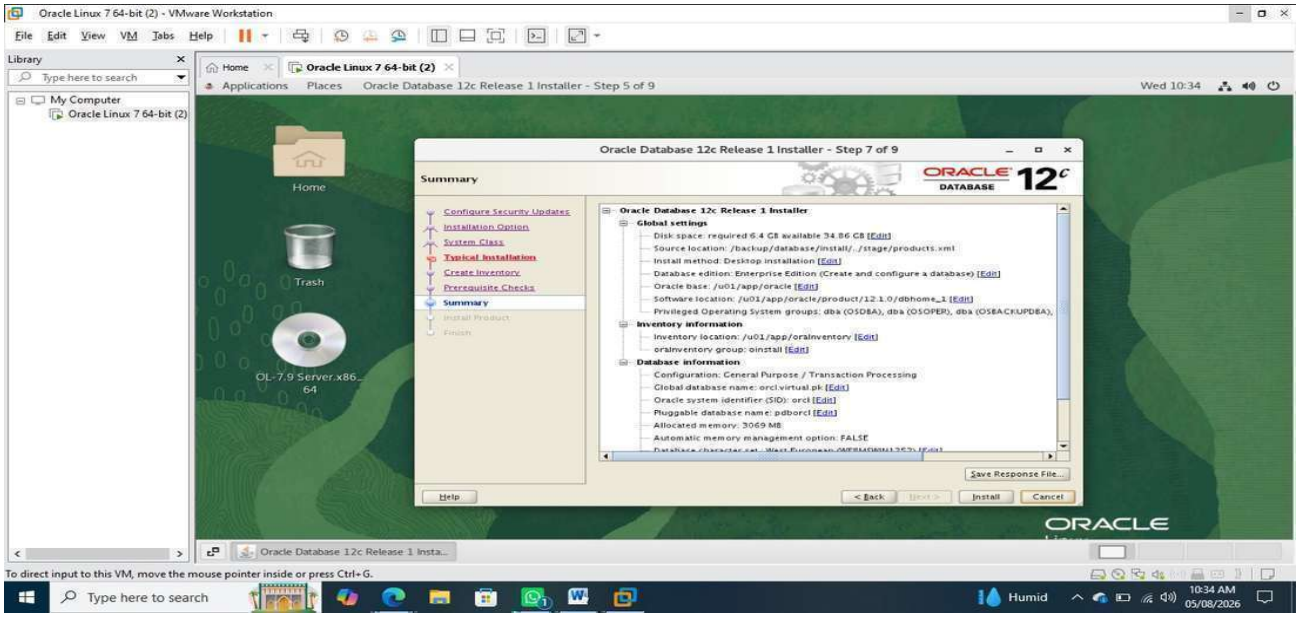
Before the installation begins, Oracle Universal Installer displays a summary of all selected configuration options. Carefully review the information to ensure it is correct.

The summary includes:

- Installation Type
- Oracle Base
- Oracle Home
- Global Database Name
- Software Location
- Operating System Groups
- Installation Components

If everything is correct, click **Install**.\

Figure 2.29: Installation Summary



Step 30: Install Oracle Database 12c

Explanation

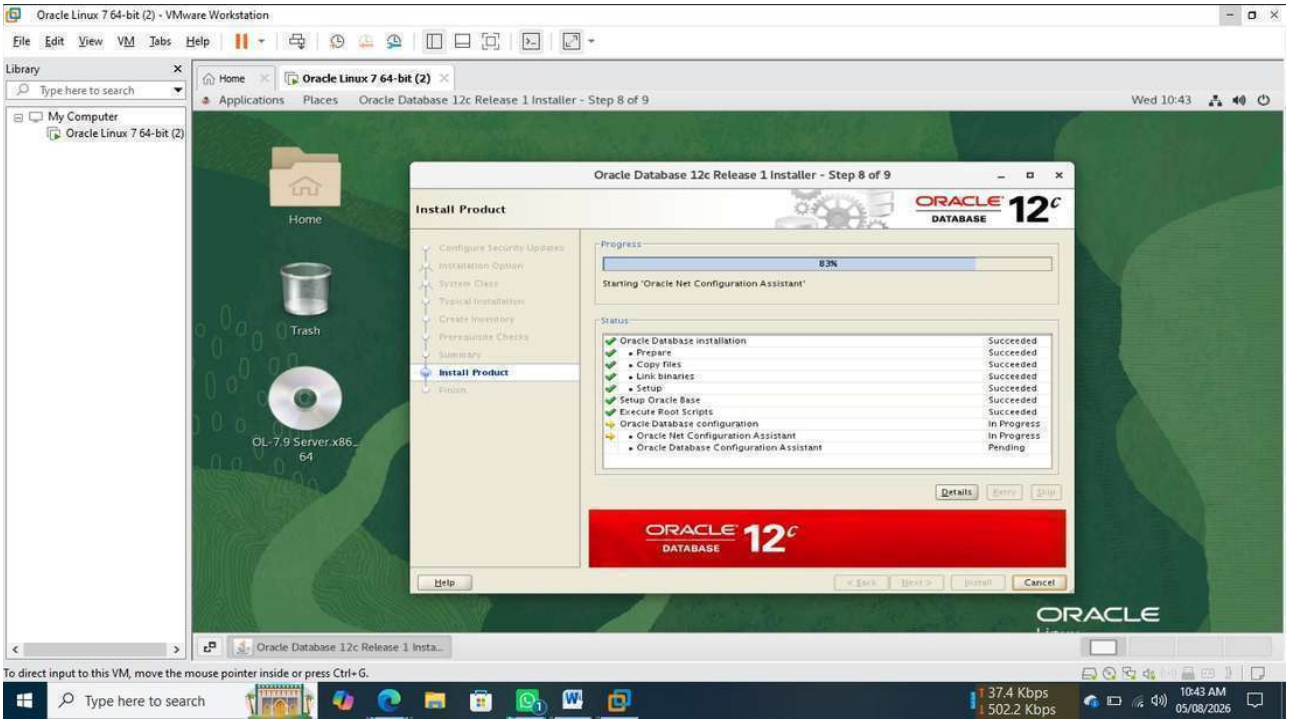
After clicking **Install**, Oracle Universal Installer begins copying files, configuring the Oracle software, and creating the Oracle Database. This process may take several minutes depending on the system performance.

Expected Result

A progress window appears showing the installation status.

Wait until the installation reaches **100%**.

Figure 2.30: Oracle Database Installation Progress



2.8 Executing Root Configuration Scripts

Introduction

During the installation, Oracle Universal Installer pauses and asks the user to execute one or more root configuration scripts. These scripts configure the Oracle Inventory and complete the operating system integration.

Step 31: Execute Root Scripts

Explanation

Open a new terminal window and log in as the **root** user. Execute the scripts exactly as displayed by the installer.

Example Commands

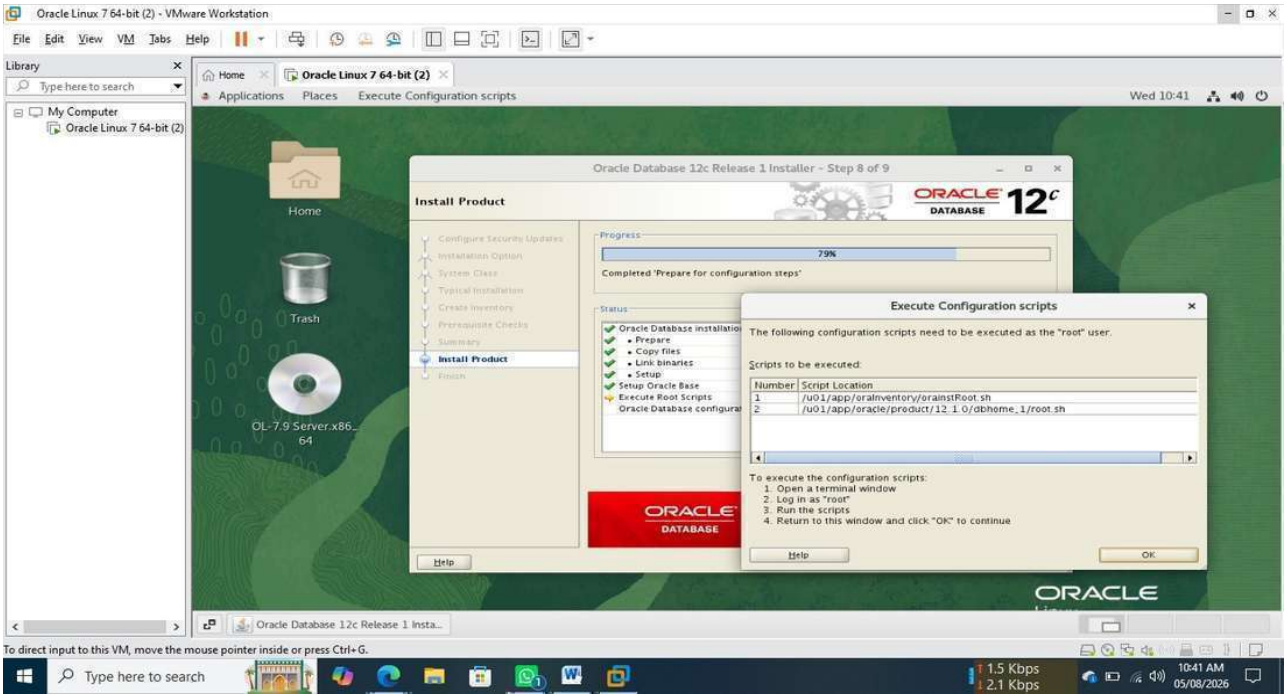
```
/u01/app/oraInventory/orainstRoot.sh
/u01/app/oracle/product/12.1.0/dbhome_1/root.sh
```

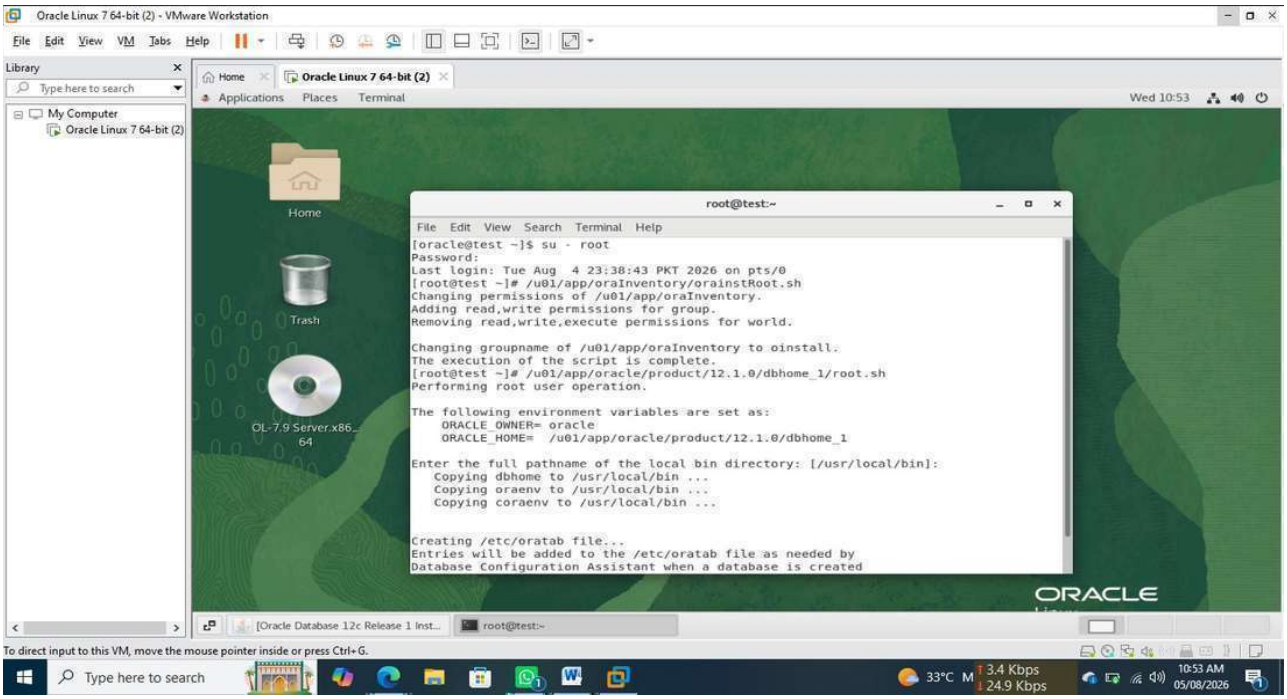
Expected Output

Both scripts should complete successfully without any errors.

After executing the scripts, return to the Oracle Universal Installer and click **OK** to continue.

Figure 2.31: Executing Root Scripts





2.9 Completing Oracle Database Installation

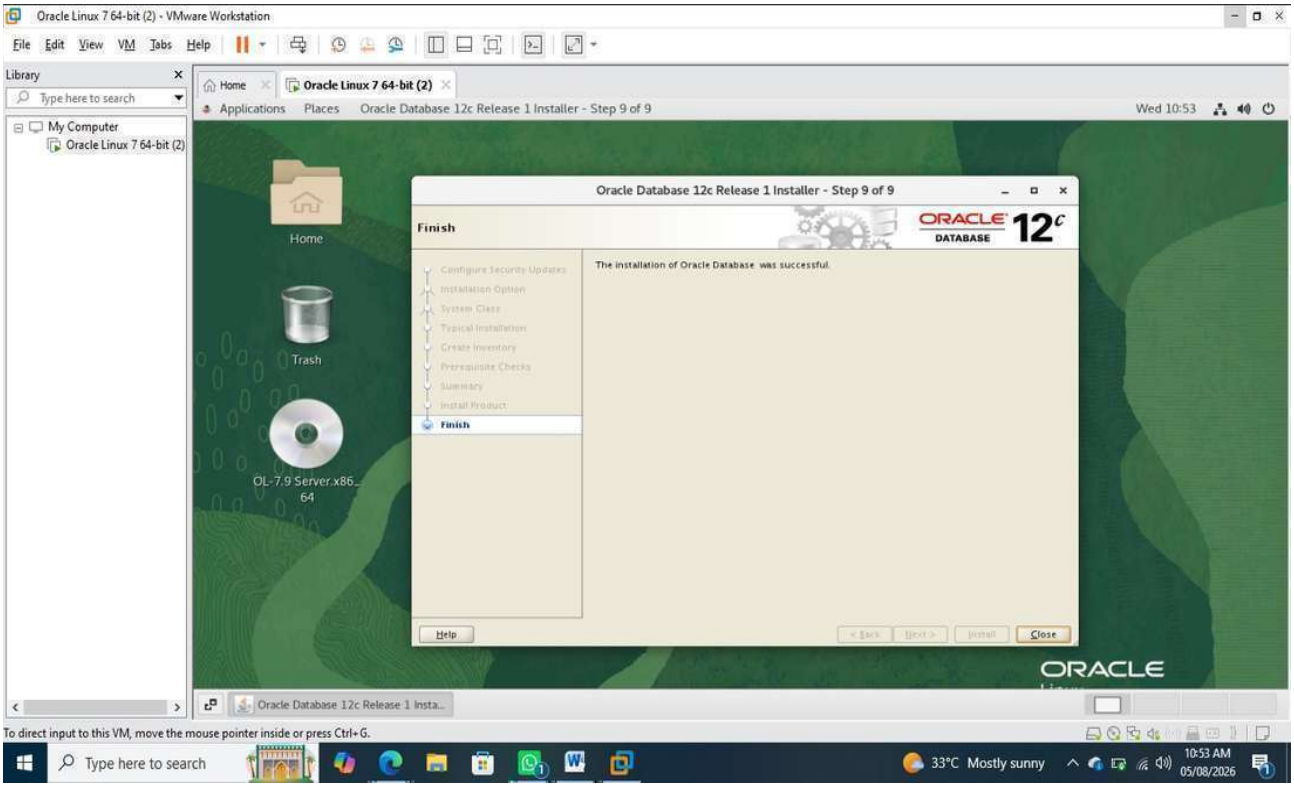
Explanation

After the root scripts have been executed successfully, Oracle Universal Installer completes the remaining configuration tasks.

When the installation finishes successfully, a confirmation message appears indicating that Oracle Database 12c has been installed successfully.

Click **Close** to exit the installer.

Figure 2.32: Oracle Database Installation Completed Successfully



CHAPTER 3 Setting Oracle Environment Variables and Starting Oracle Database 12c After Reboot

3.1 Introduction

After restarting Oracle Linux, the Oracle Database services may not start automatically, depending on the system configuration. Before accessing the database, the Oracle environment must be configured correctly by setting the required environment variables. Once the environment is configured, the Oracle Net Listener service is started, followed by the Oracle Database instance. Finally, the database is verified using SQL*Plus.

This practical provides hands-on experience in performing the routine startup procedure that every Oracle Database Administrator (DBA) should know.

3.2 Objective

The objectives of this practical were:

- To log in as the Oracle software owner.
- To configure Oracle environment variables.
- To verify the Oracle environment.
- To start the Oracle Net Listener.
- To verify the Listener status.
- To start the Oracle Database instance.
- To verify that the database is running successfully.
- To connect to Oracle Database using SQL*Plus.

3.3 System Requirements

Component	Description
Operating System	Oracle Linux 7.9
Database	Oracle Database 12c
User	oracle
Terminal	Oracle Linux Terminal
SQL Utility	SQL*Plus

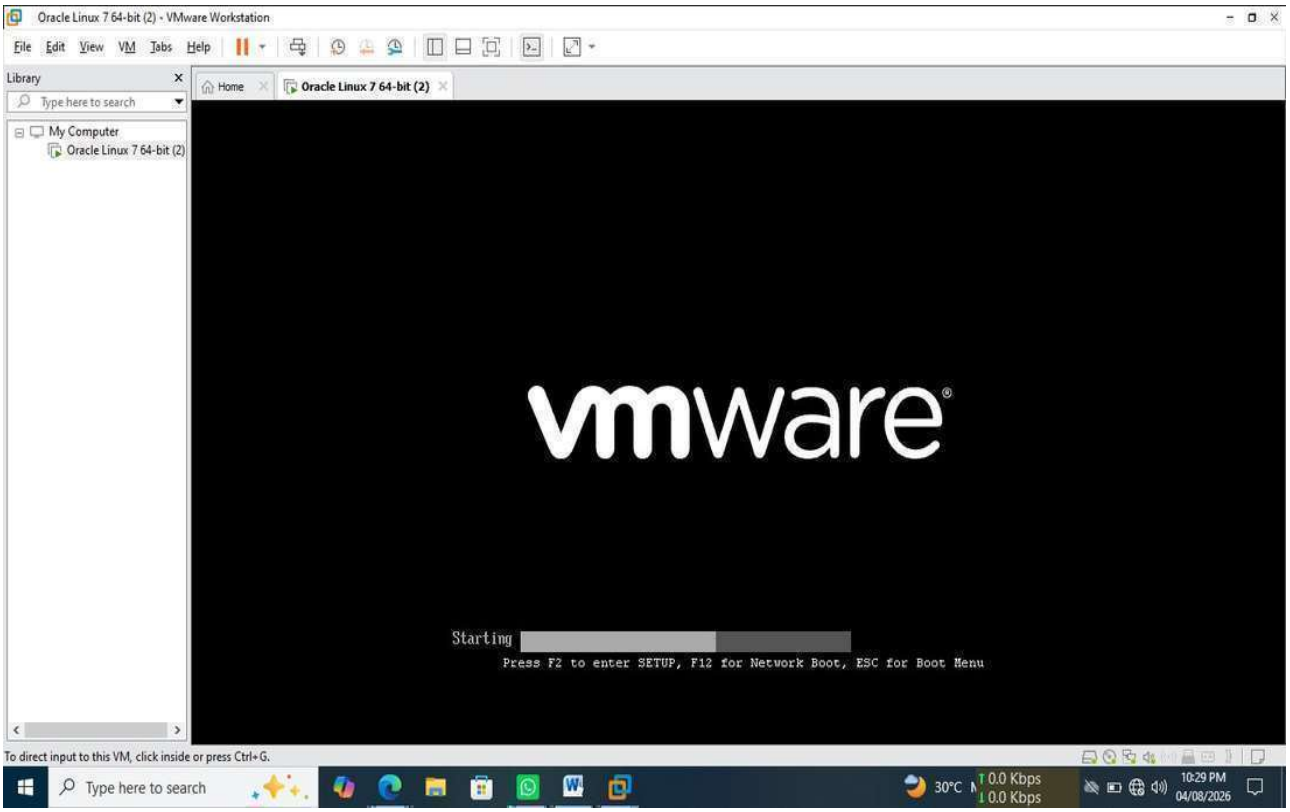
Step 1: Start Oracle Linux

Power on the virtual machine from VMware Workstation and wait until Oracle Linux boots successfully.

Explanation

Oracle Linux must be fully started before any Oracle services can be accessed.

Figure 3.1: Oracle Linux Boot Screen



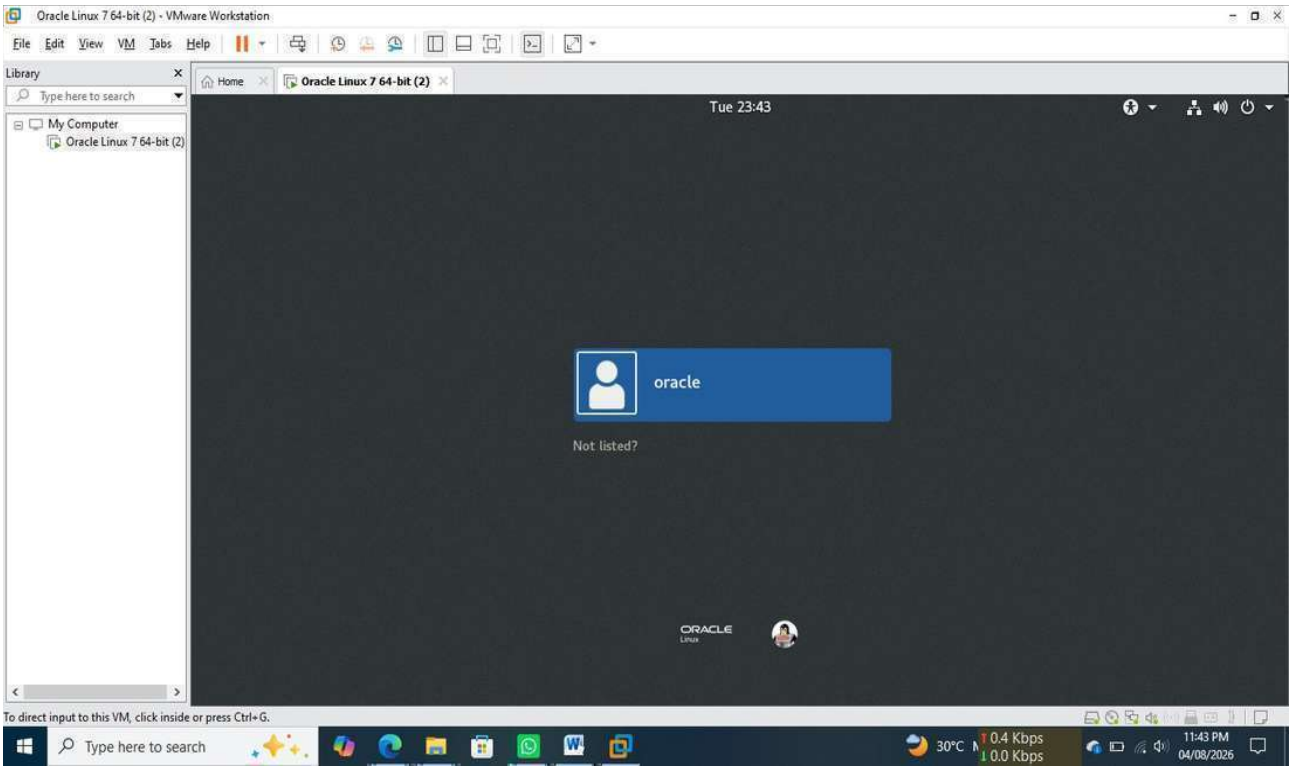
Step 2: Log in as the Oracle User

Log in using the **oracle** user account.

Explanation

The Oracle user is the owner of the Oracle software installation and is responsible for managing the Oracle Database.

Figure 3.2: Oracle User Login



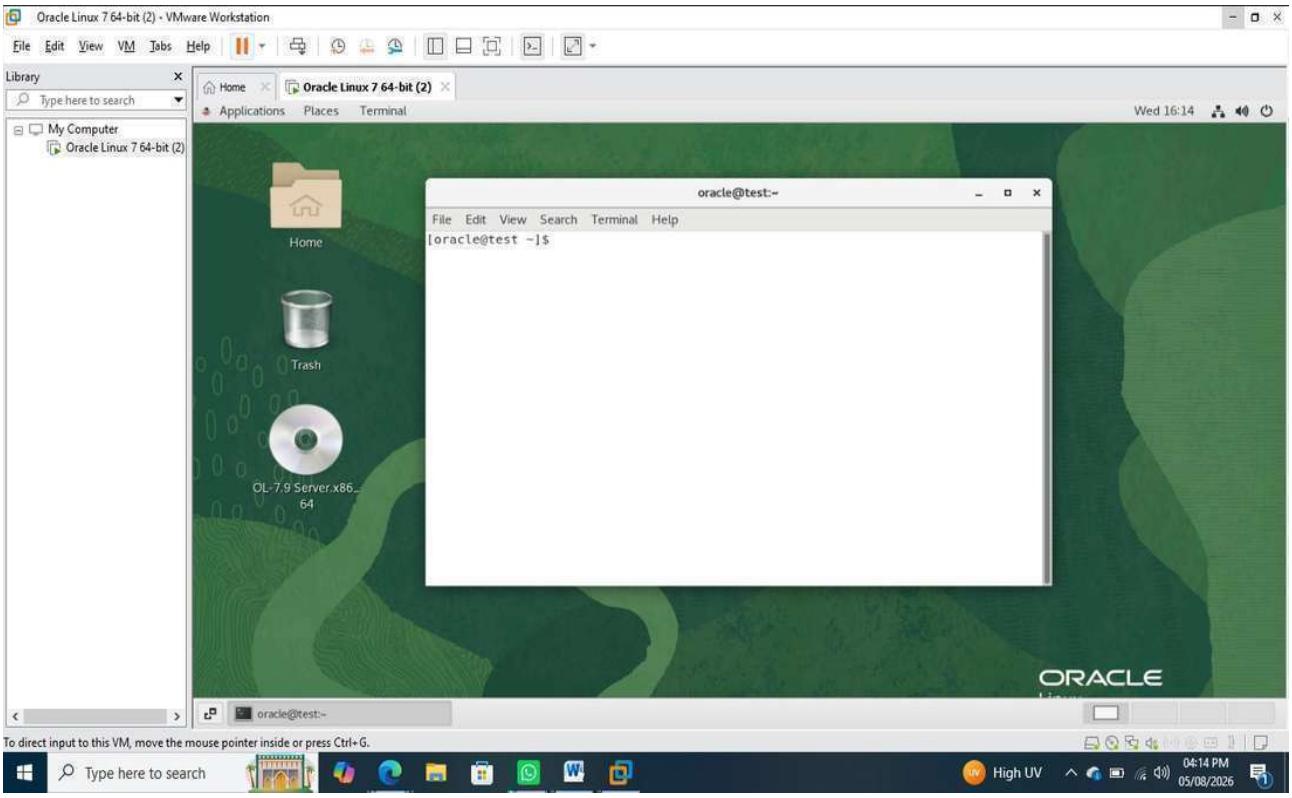
Step 3: Open the Terminal

After logging in, open the Terminal application.

Explanation

All Oracle startup commands are executed through the Linux Terminal.

Figure 3.3: Oracle Linux Terminal



Step 4: Verify the Current User

Before executing Oracle commands, verify the logged-in user.

Command

whoami

Expected Output

oracle

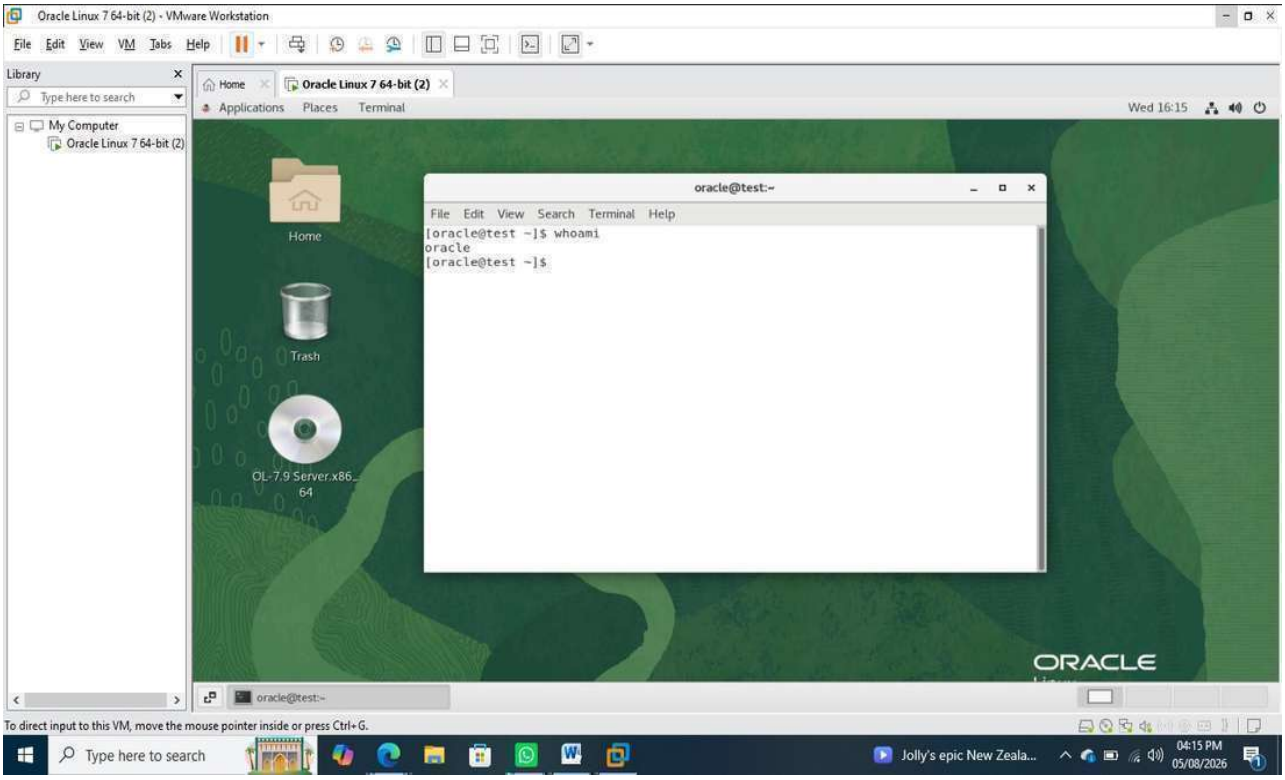
Command Explanation

- whoami displays the name of the currently logged-in user.

Explanation

This confirms that you are logged in as the Oracle software owner.

Figure 3.4: Verifying Oracle User



Step 5 : Configure Oracle Environment Using oraenv

If the environment variables are not available after reboot, configure them automatically.

Commands

. oraenv

Command Explanation

The system may display:

ORACLE_SID = [oracle] ?

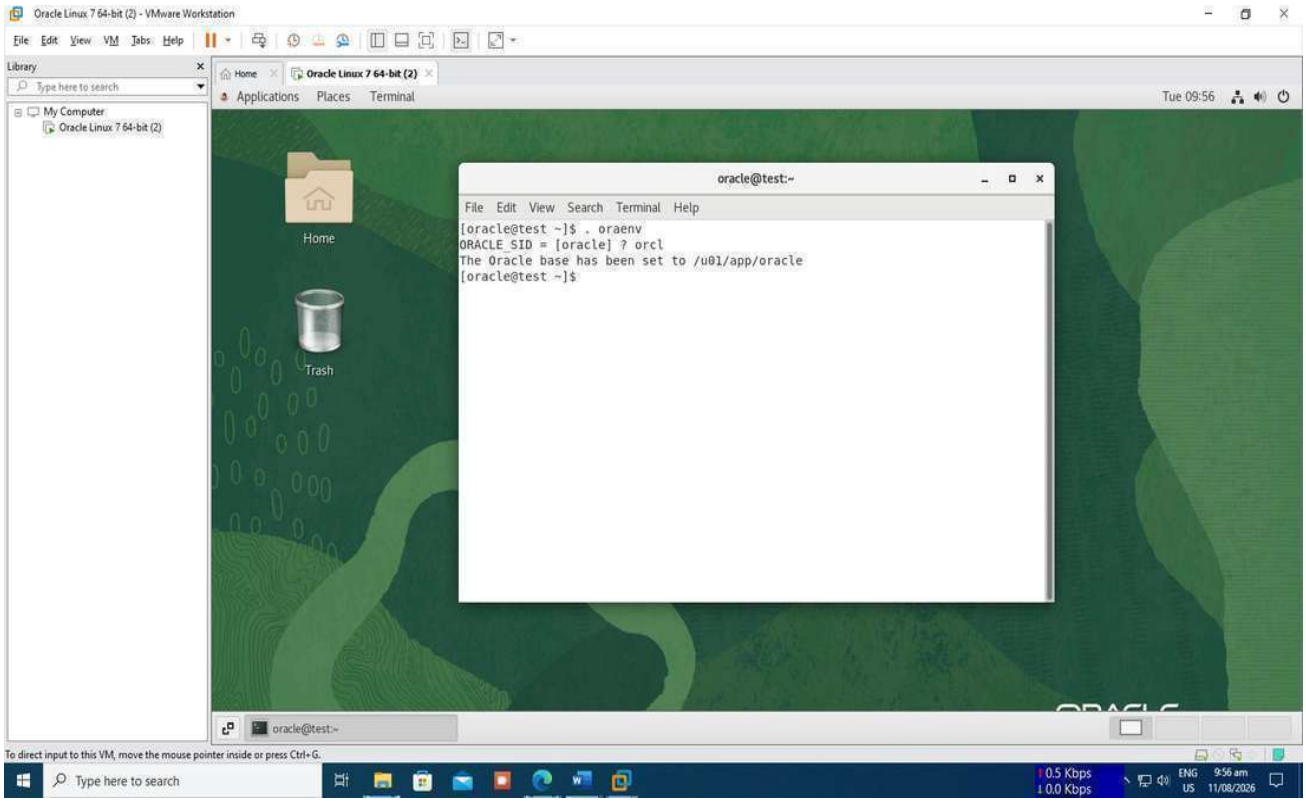
Enter : orcl

After entering the SID, oraenv reads the Oracle configuration and sets the appropriate environment variables.

Explanation

The oraenv utility is used to configure the Oracle environment automatically. It obtains the Oracle Home information from the Oracle configuration file /etc/oratab and sets the required environment variables for the selected Oracle SID.

Figure 3.5 : Configure Oracle Environment Variables



Step 6 : Check the Oracle Environment Variables

Instead of checking each variable separately, the following commands can also be executed

Commands

```
echo $ORACLE_SID
echo $ORACLE_HOME
echo $PATH
```

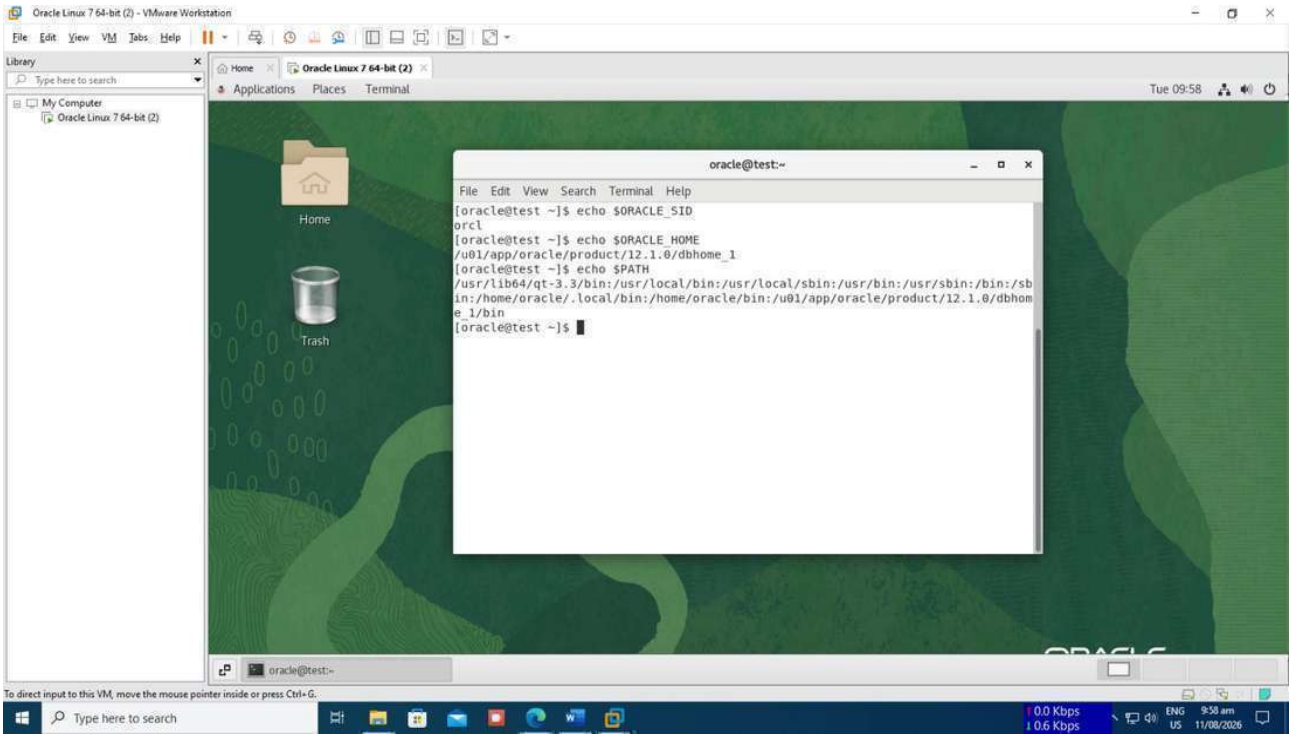
Expected Output (Example)

```
orcl
/u01/app/oracle/product/12.1.0/dbhome_1
/usr/lib64/qt-/bin:/usr/local/bin:/usr/local/sbin:/usr/bin:/usr/sbin:/bin:/sbin:/home/
oracle/.local/bin:/home/oracle/bin:/u01/app/oracle/product/12.1.0/dbhome_1/bin
```

Explanation

These commands verify that the major Oracle environment variables have been configured successfully by oraenv.

Figure 3.6 : Oracle Environment Variables



Starting Oracle Net Listener and Oracle Database

After configuring the Oracle environment variables, the next step is to start the Oracle Net Listener service and then start the Oracle Database instance using SQL*Plus.

Step 7 : Start the Oracle Net Listener

Start the Oracle Net Listener service.

Command

lsnrctl start

Command Explanation

- lsnrctl → Oracle Listener Control Utility.
- start → Starts the Oracle Net Listener service.

Expected Output

Starting /u01/app/oracle/product/12.1.0/dbhome_1/bin/tnslsnr...

TNSLSNR for Linux: Version 12.1.0.2.0

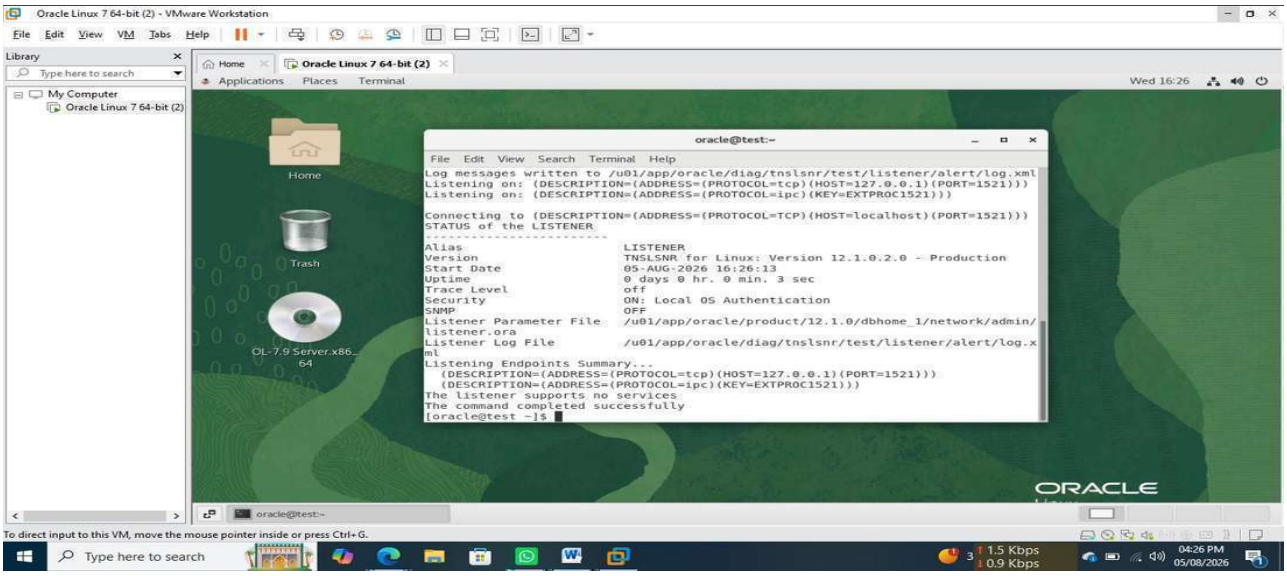
The command completed successfully.

Explanation

The Oracle Net Listener accepts incoming connection requests from client applications such as SQL Developer and TOAD.

Without the Listener service, remote database connections cannot be established.

Figure 3.7: Starting Oracle Net Listener



Step 8: Verify Listener Status

Check whether the Listener is running successfully.

Command

lsnrctl status

Command Explanation

- status displays the current status of the Oracle Net Listener.

Expected Output

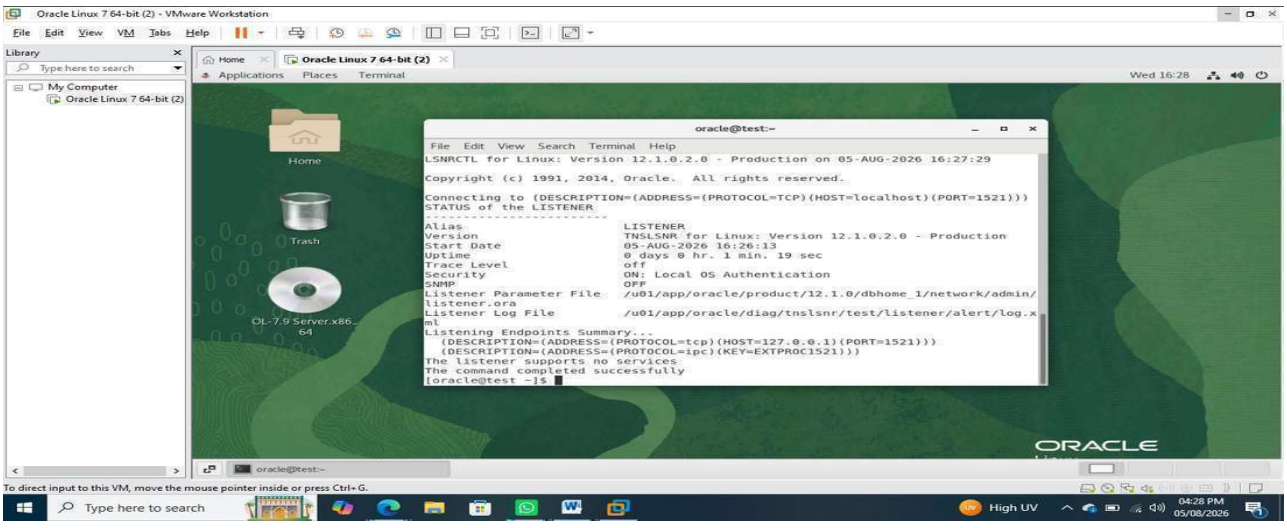
The output should display:

- Listener Name
- Version
- Listening Endpoints
- Services Summary
- Status: READY

Explanation

This command confirms that the Listener service has started successfully and is ready to accept client connections.

Figure 3.8: Oracle Listener Status



Step 9: Open SQL*Plus

Log in to SQL*Plus as the SYSDBA user.

Command

sqlplus / as sysdba

Command Explanation

- sqlplus → Starts SQL*Plus.
- / → Uses Operating System Authentication.
- as sysdba → Connects with Database Administrator privileges.

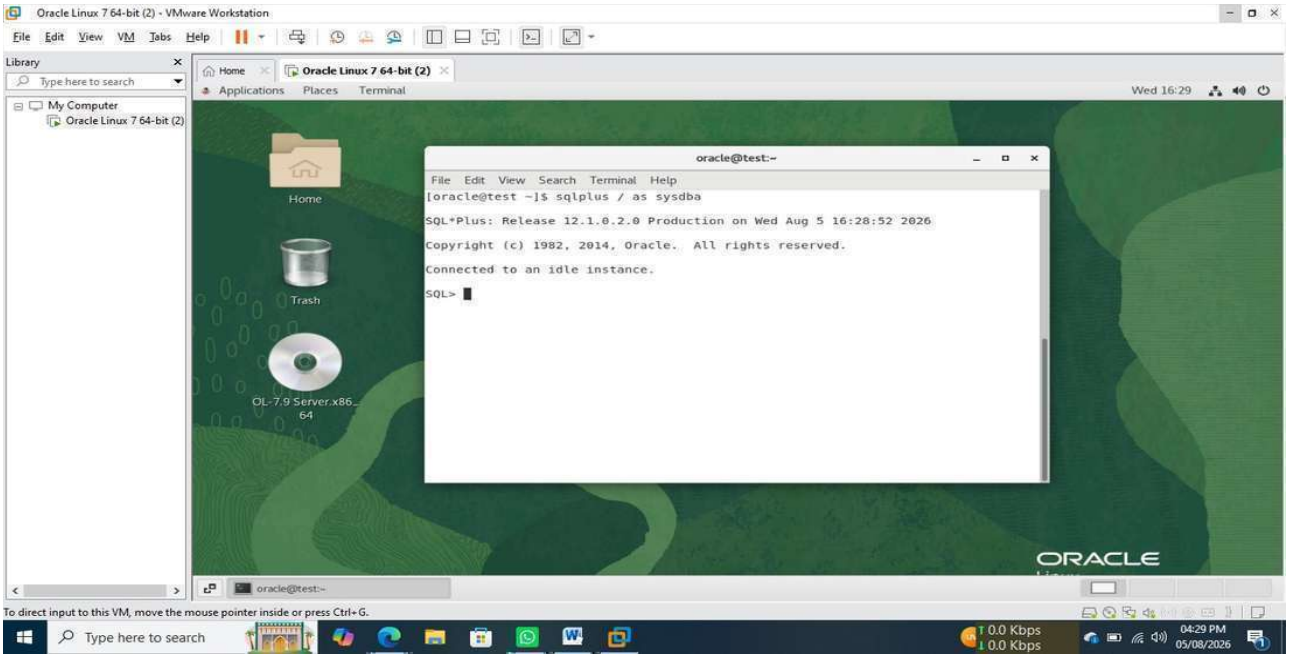
Expected Output

SQL>

Explanation

SYSDBA is the highest administrative privilege in Oracle Database and is used to start, stop, and manage the database.

Figure 3.9: Logging into SQL*Plus



Step 10: Check the Database Instance Status

Before starting the database, check its current status.

Command

SELECT STATUS FROM V$INSTANCE;

Expected Output

If the database is not started:

```
STATUS
------------
STARTED
```

or

```
STATUS
------------
MOUNTED
```

If the database is already open:

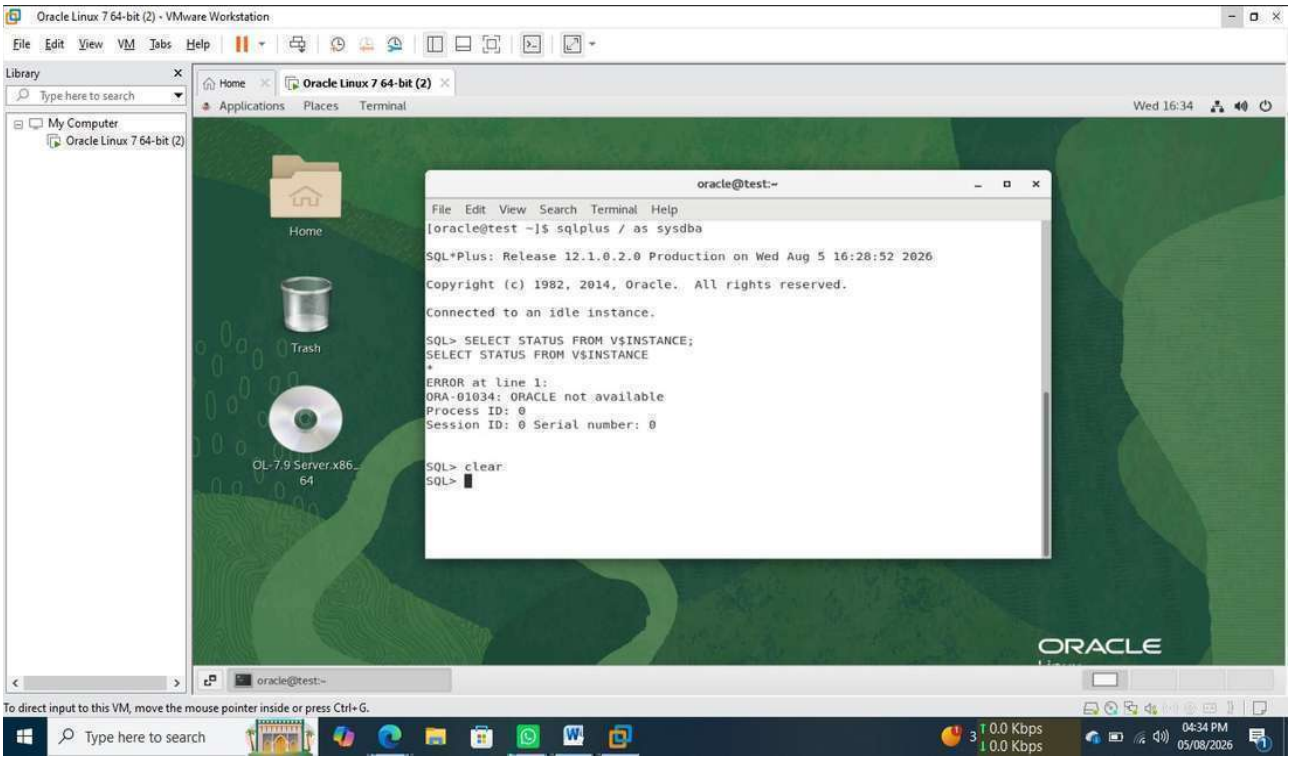
STATUS

OPEN

Explanation

This command displays the current state of the Oracle instance.

Figure 3.10: Checking Database Instance Status



Step 11: Start the Oracle Database

If the database is not already open, start it.

Command

STARTUP;

Command Explanation

The STARTUP command:

- Starts the Oracle instance.
- Mounts the database.
- Opens the database for users.

Expected Output

ORACLE instance started.

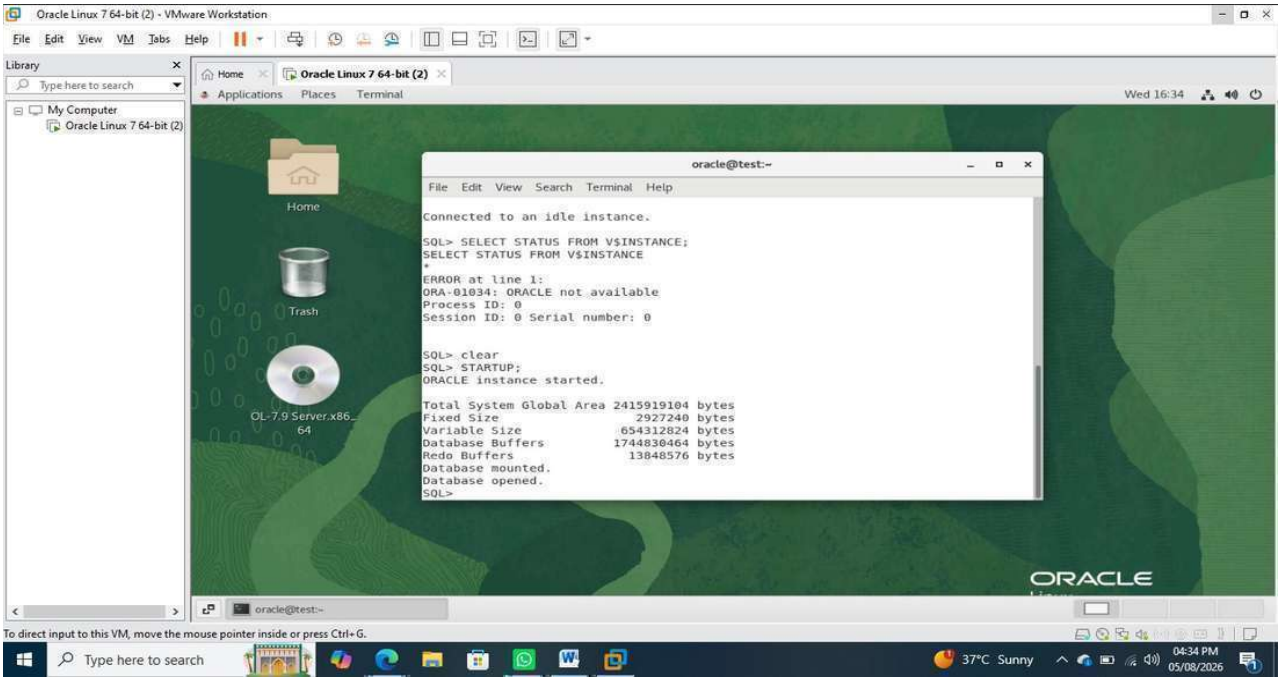
Database mounted.

Database opened.

Explanation

This is one of the most frequently used commands by Oracle Database Administrators after a server restart.

Figure 3.11: Starting Oracle Database



Step 12: Verify the Database Name

Check the name of the Oracle Database.

Command

SELECT NAME FROM V$DATABASE;

Expected Output

NAME

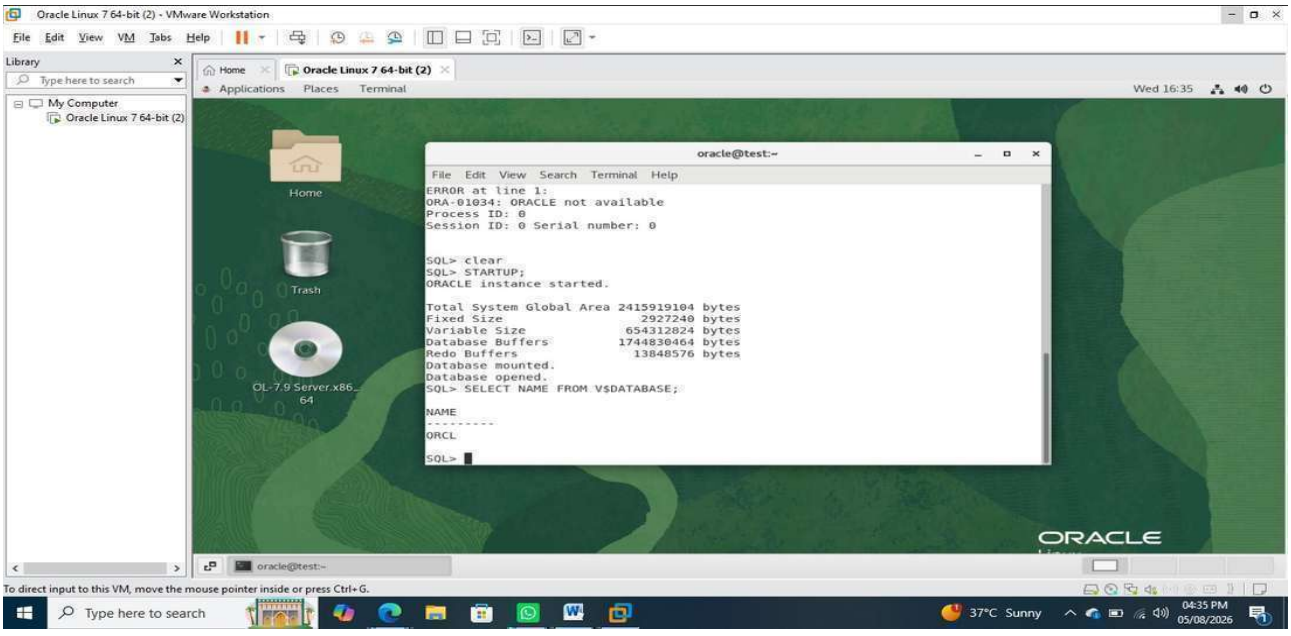
ORCL

(Replace ORCL with your actual database name if it is different.)

Explanation

This command confirms the identity of the connected Oracle database.

Figure 3.12: Verifying Database Name



Step 13: Verify the Database Open Mode

Check whether the database is open.

Command

SELECT OPEN_MODE FROM V$DATABASE;

Expected Output

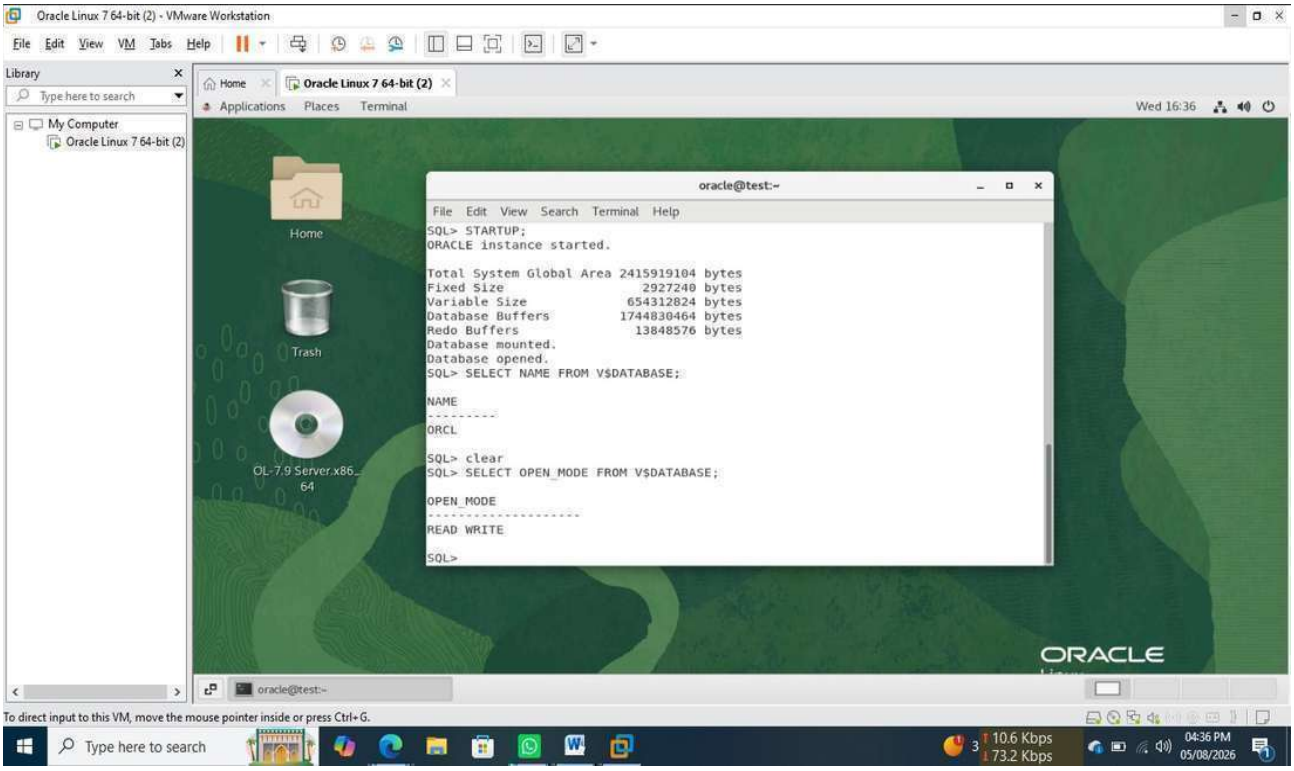
OPEN_MODE

READ WRITE

Explanation

The READ WRITE mode indicates that the database is fully operational and users can perform both read and write operations.

Figure 3.13: Checking Database Open Mode



Step 14: Display the Current Database User

Verify the currently logged-in user.

Command

SHOW USER;

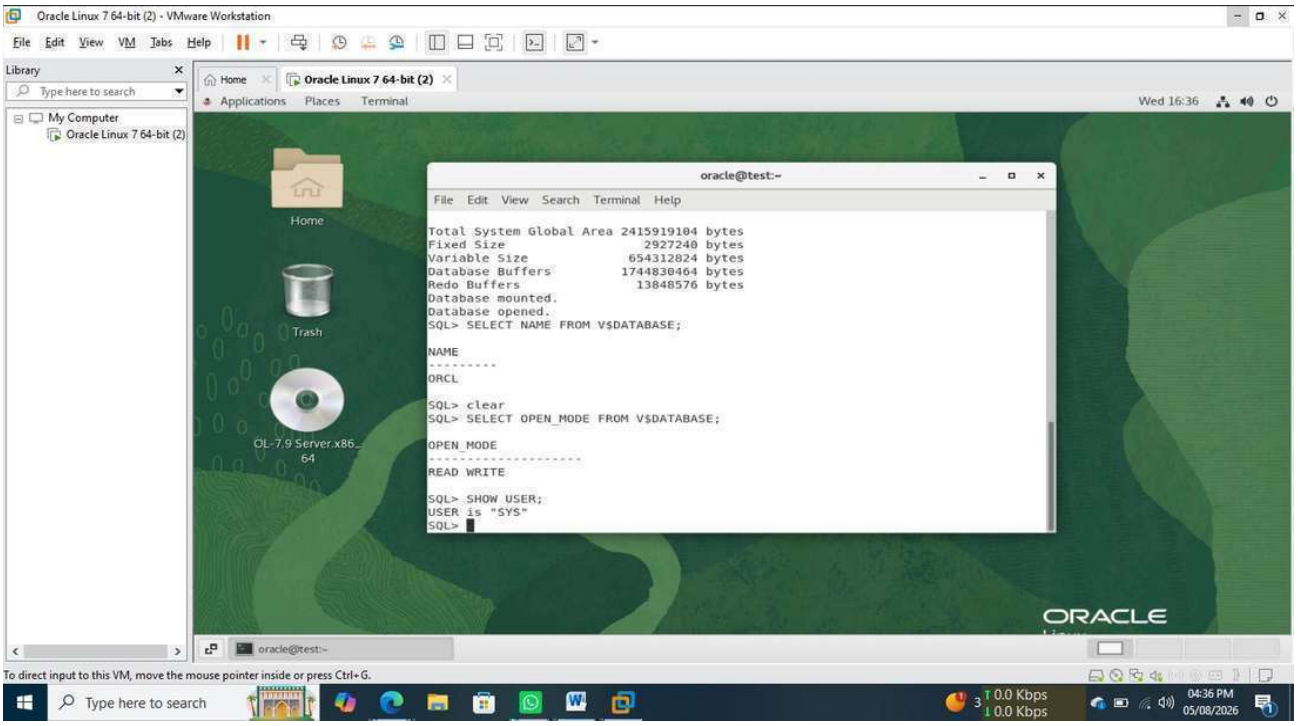
Expected Output

USER is "SYS"

Explanation

This command confirms that the session is connected as the SYS user with administrative privileges.

Figure 3.14: Current Database User



Step 15: Exit SQL*Plus

After completing the verification, exit SQL*Plus.

Command

EXIT;

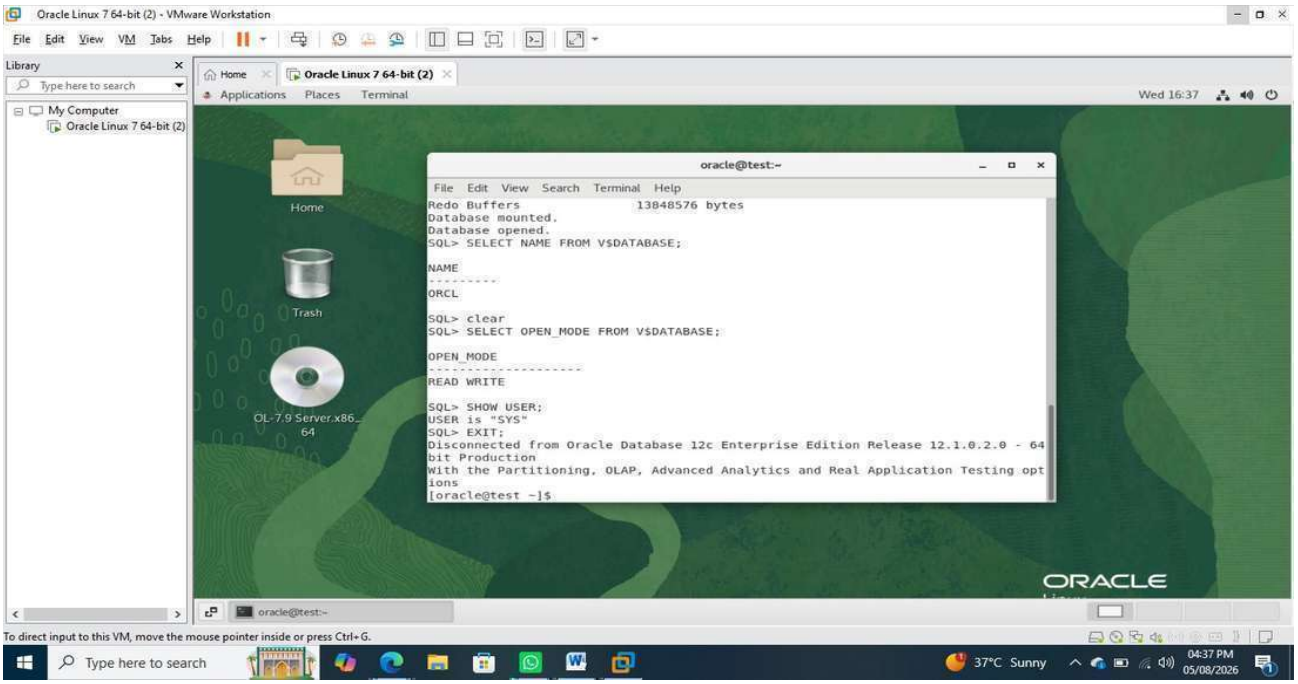
Command Explanation

The EXIT command closes the SQL*Plus session and returns to the Linux terminal.

Explanation

Always exit SQL*Plus properly after completing administrative tasks.

Figure 3.15: Exiting SQL*Plus



Key Commands Used in This Practical

Command	Purpose
whoami	Verify the current Linux user
echo $ORACLE_HOME	Display Oracle Home directory
echo $ORACLE_SID	Display the Oracle SID
export ORACLE_HOME=...	Set Oracle Home environment variable
export ORACLE_BASE=...	Set Oracle Base environment variable
export ORACLE_SID=orcl	Set the Oracle SID
export PATH=$ORACLE_HOME/bin:$PATH	Add Oracle binaries to the system PATH
lsnrctl start	Start the Oracle Net Listener
lsnrctl status	Check the Listener status
sqlplus / as sysdba	Connect to Oracle as SYSDBA
STARTUP;	Start the Oracle Database
SELECT STATUS FROM V$INSTANCE;	Check the instance status
SELECT NAME FROM V$DATABASE;	Display the database name
SELECT OPEN_MODE FROM V$DATABASE;	Check the database open mode
SHOW USER;	Display the current Oracle user

EXIT; Exit SQL*Plus

***** Best of Luck *****